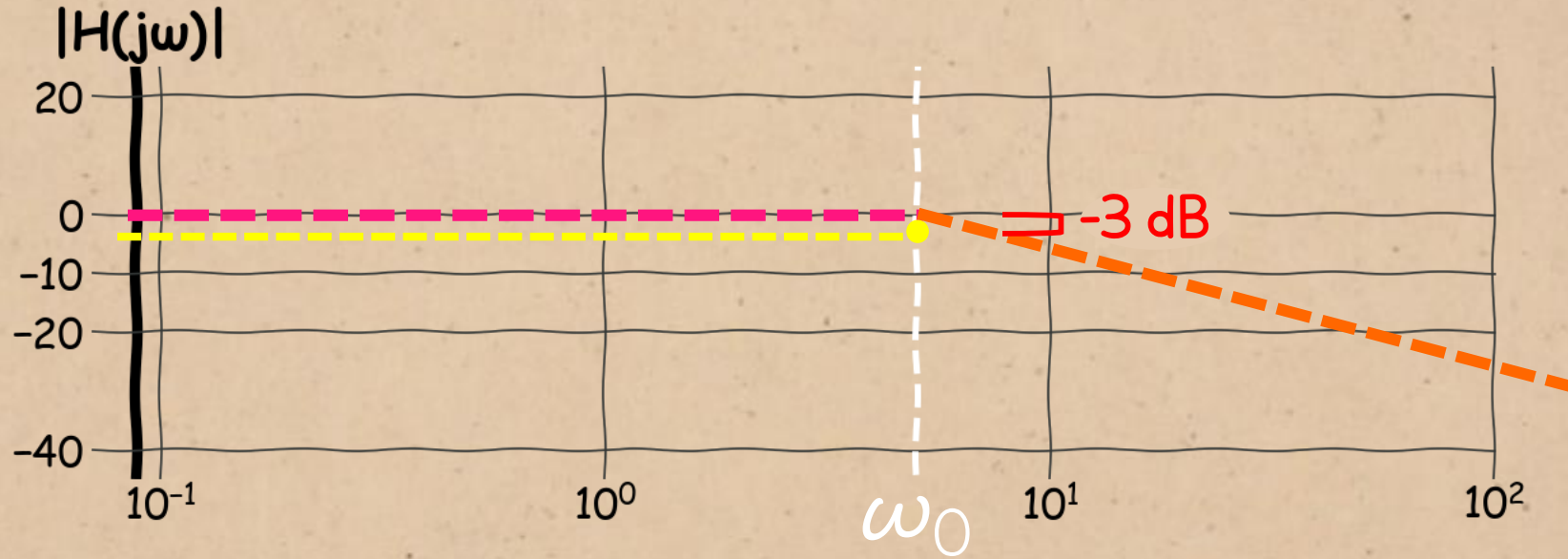




CHAPTER 8:

COMPENSATION

APPROXIMATING THE TIME CONSTANT
IN THE FREQUENCY DOMAIN



UCT
MECHATRONICS



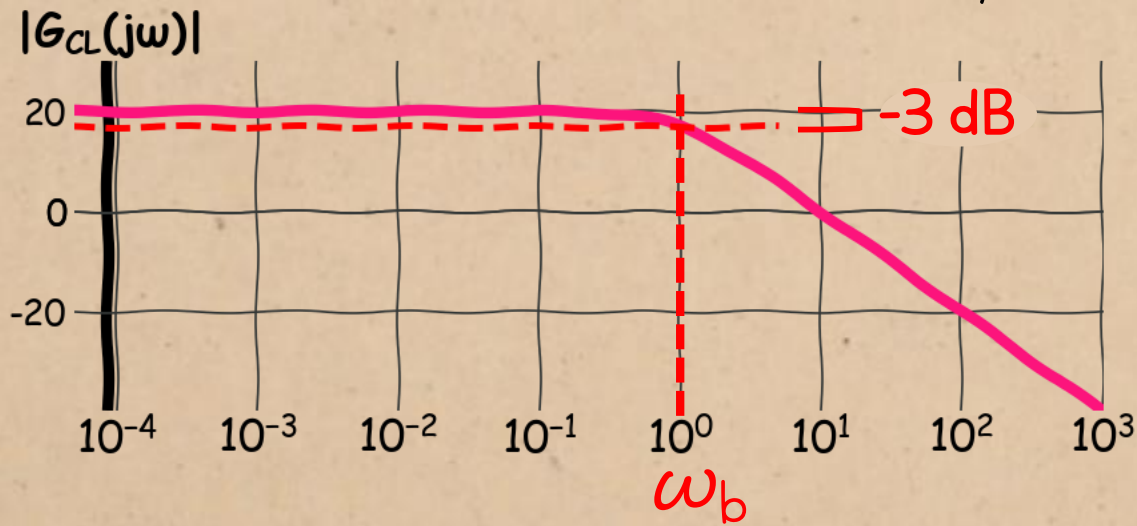
CHAPTER 8

COMPENSATION

TIME RESPONSE DESIGN IN THE
FREQUENCY DOMAIN

BANDWIDTH

In the previous lecture, I said that we can approximate the time constant by finding the -3 dB bandwidth of the system. This is true, but only for first-order systems.



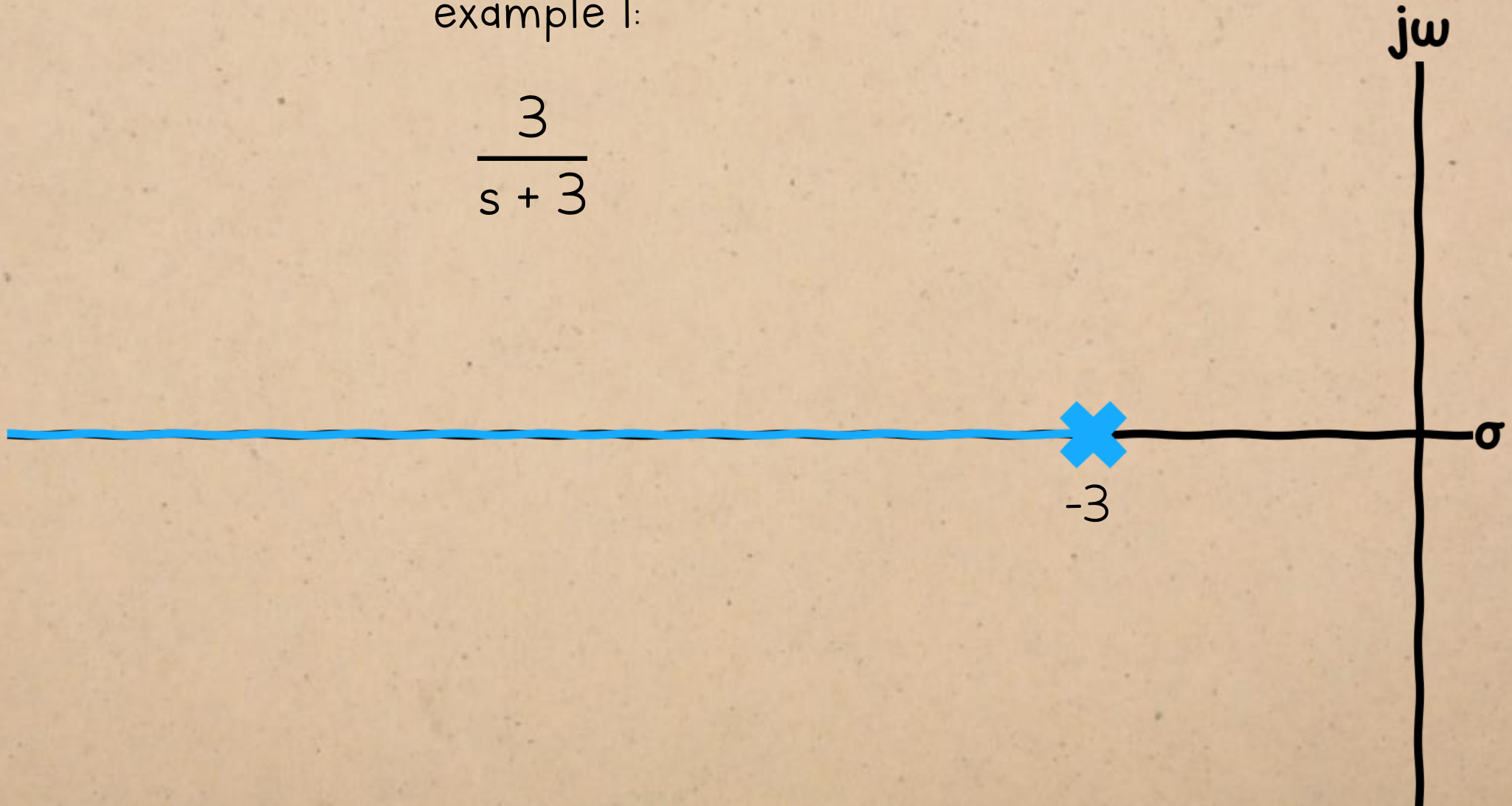
-3 dB bandwidth = the frequency where the magnitude drops to 3 decibels below the peak value.

1. Accuracy (steady-state error) depends on the DC gain
2. Speed (time constant). depends on the bandwidth
if the system is first-order
3. Damping (damping ratio)

FIRST-ORDER SYSTEM

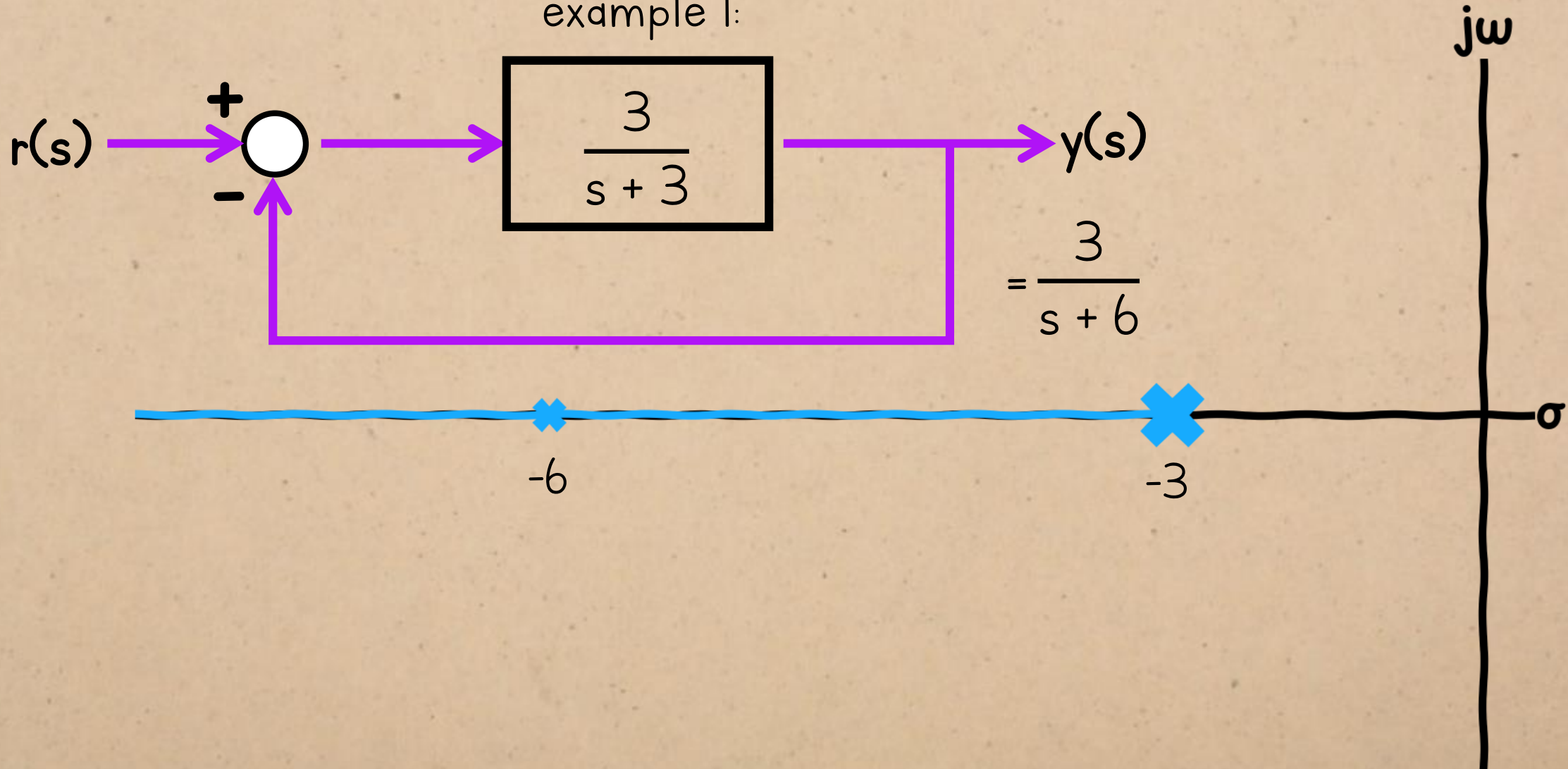
example 1:

$$\frac{3}{s + 3}$$



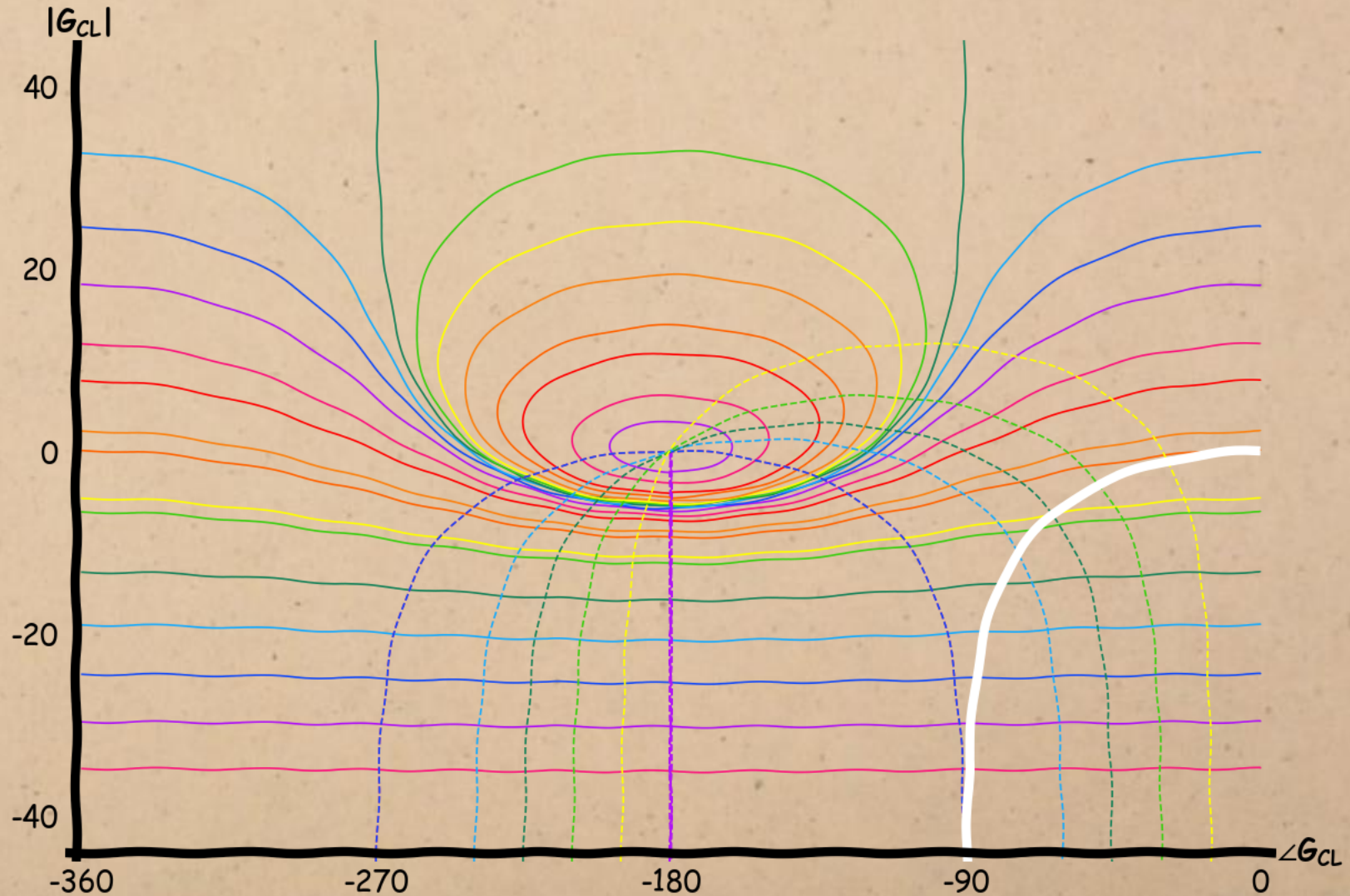
FIRST-ORDER SYSTEM

example 1:



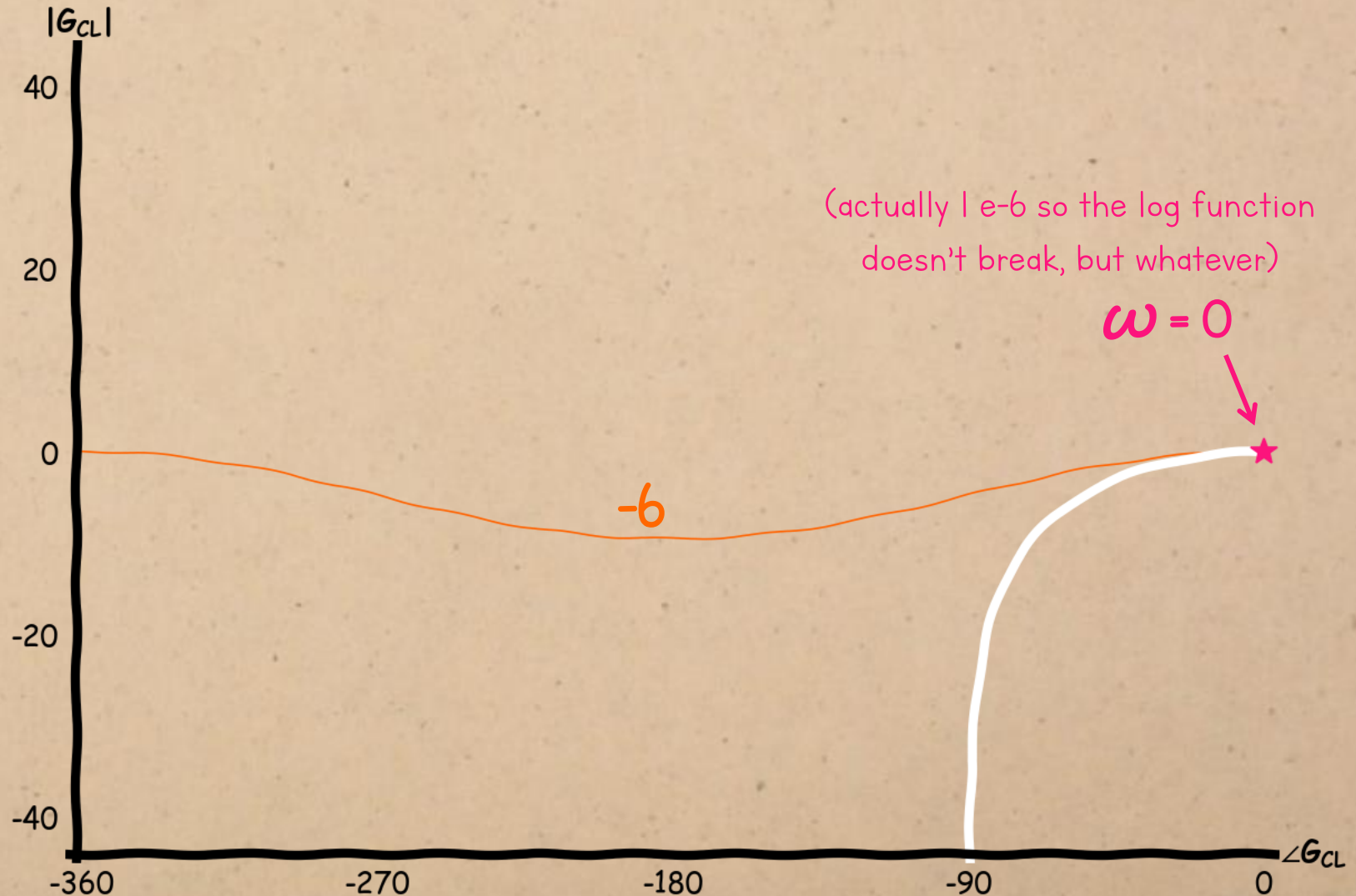
FIRST-ORDER SYSTEM

This is the Nichols chart
of the open-loop
frequency response.



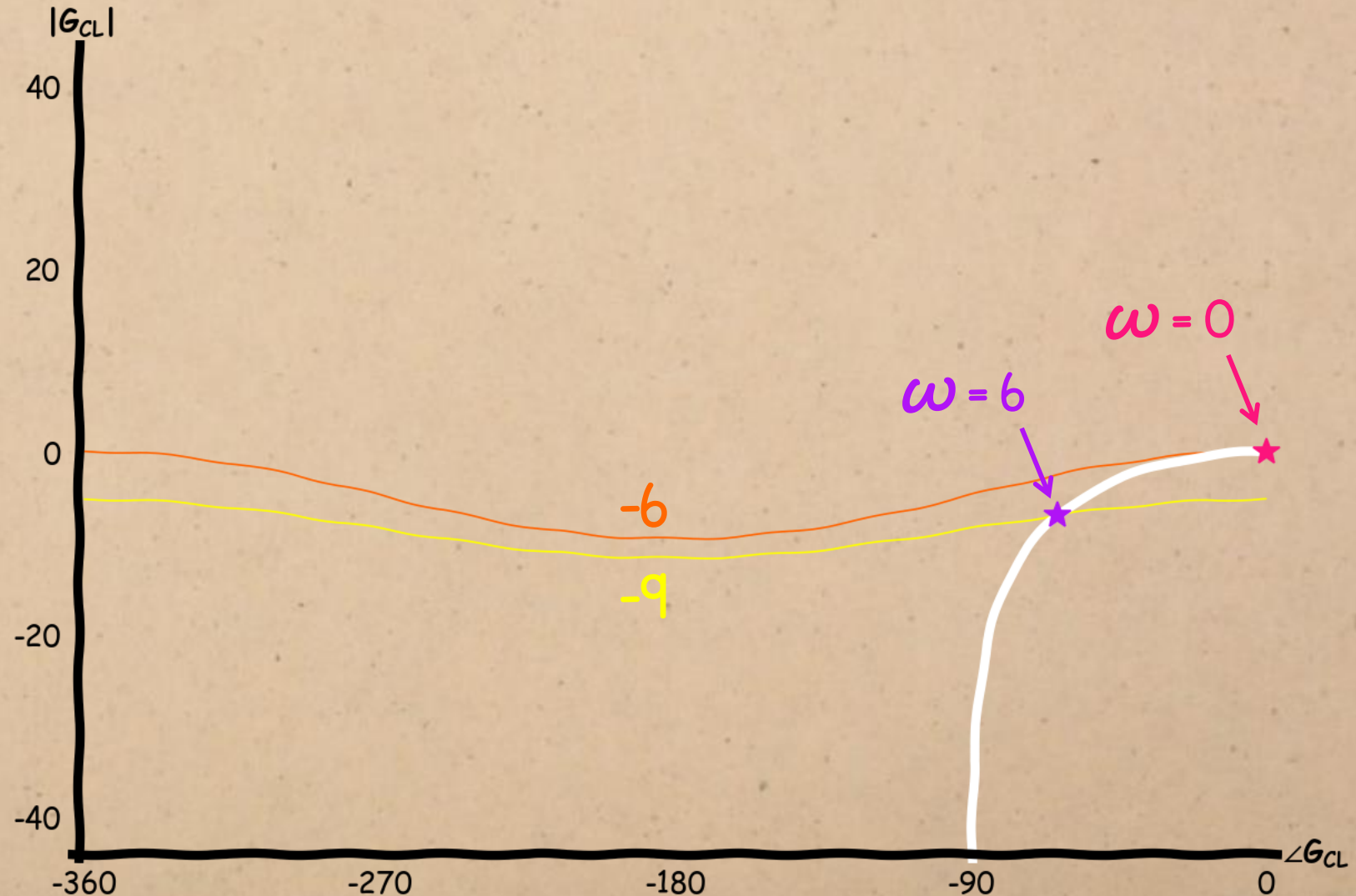
FIRST-ORDER SYSTEM

The DC gain of the closed-loop system is -6 decibels.

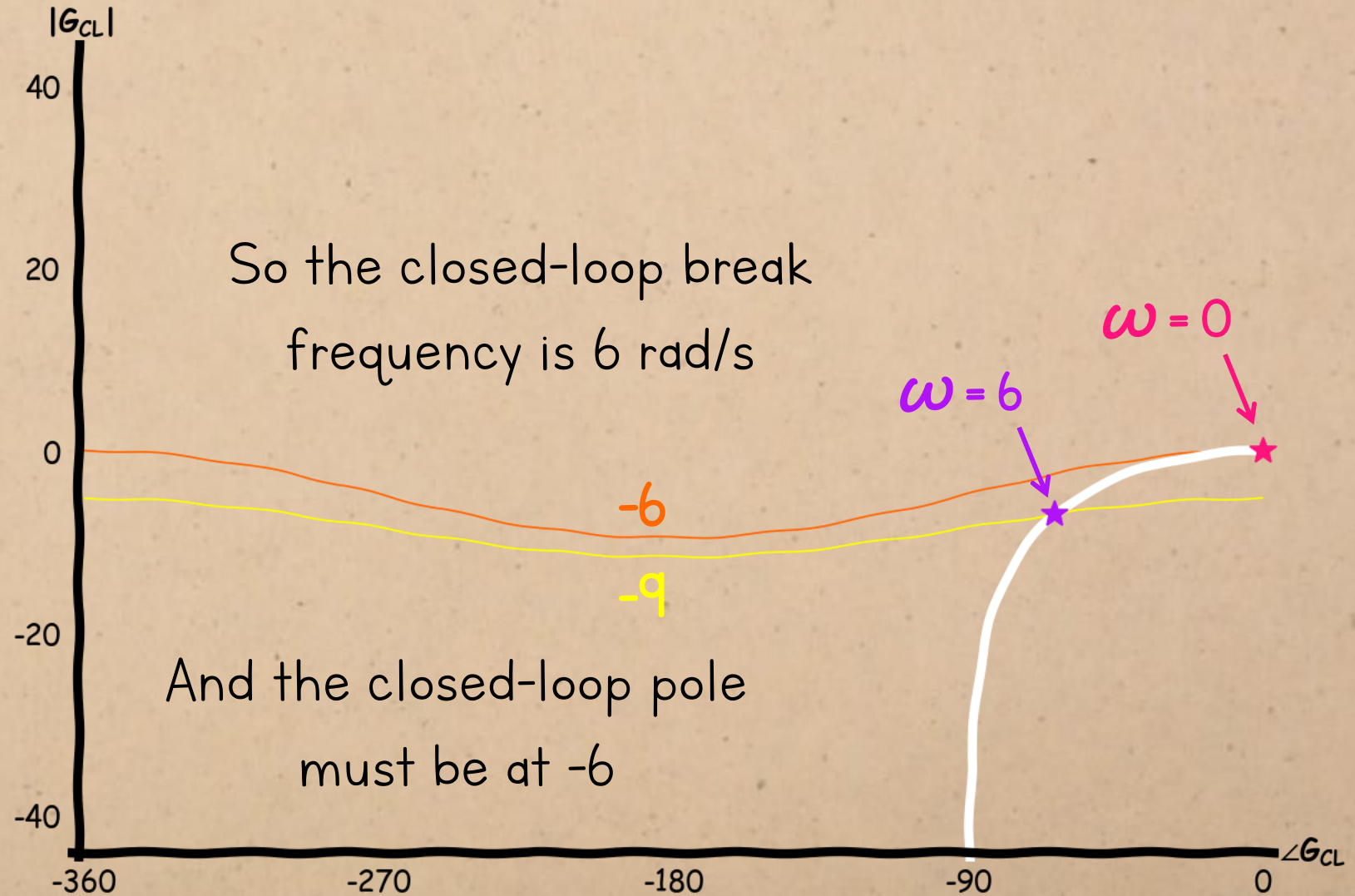


FIRST-ORDER SYSTEM

∴ to find the -3 dB bandwidth, we want to find where it intersects the -9 dB circle.

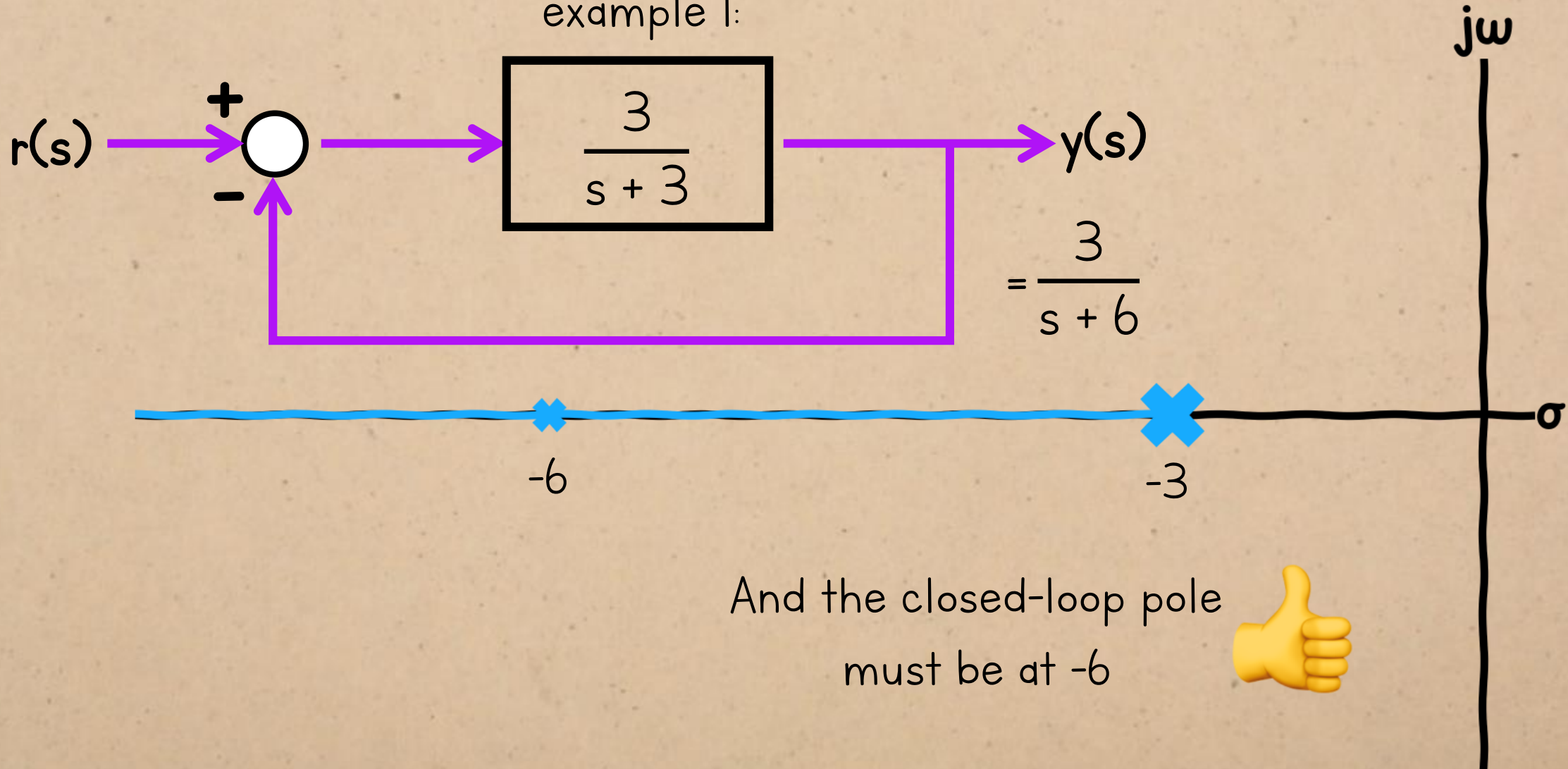


FIRST-ORDER SYSTEM



FIRST-ORDER SYSTEM

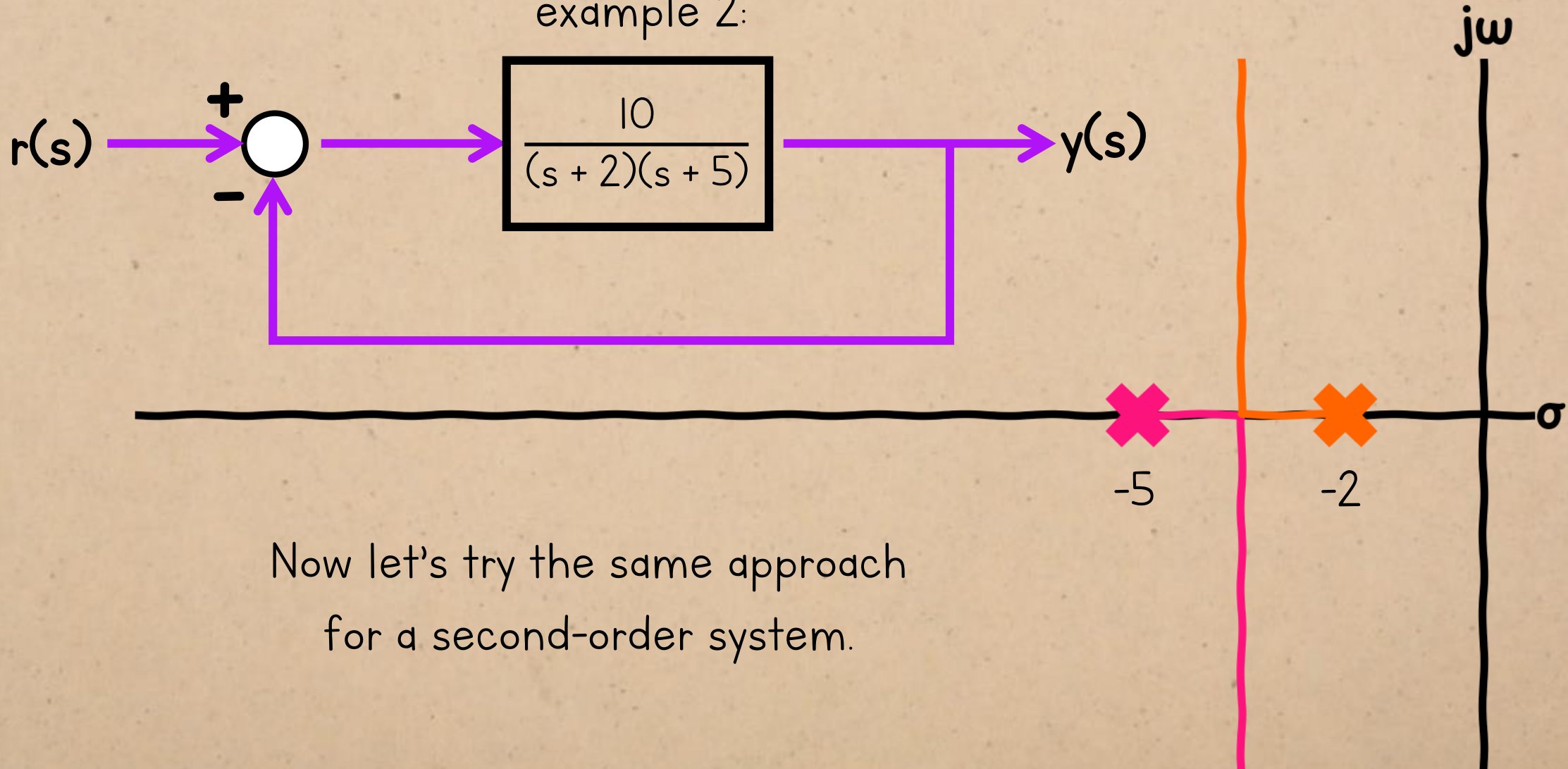
example 1:



And the closed-loop pole
must be at -6 👍

SECOND-ORDER SYSTEM

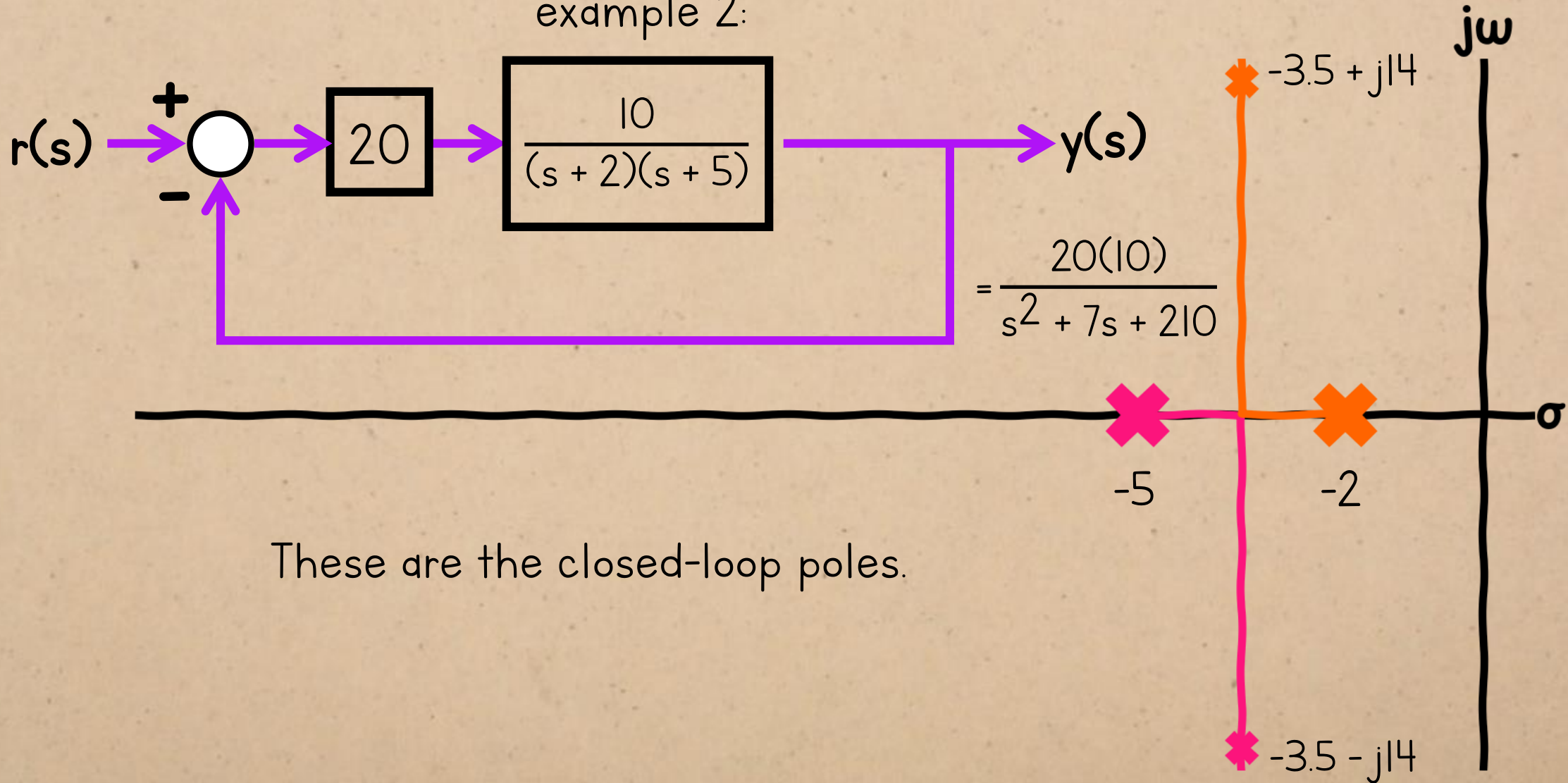
example 2:



Now let's try the same approach
for a second-order system.

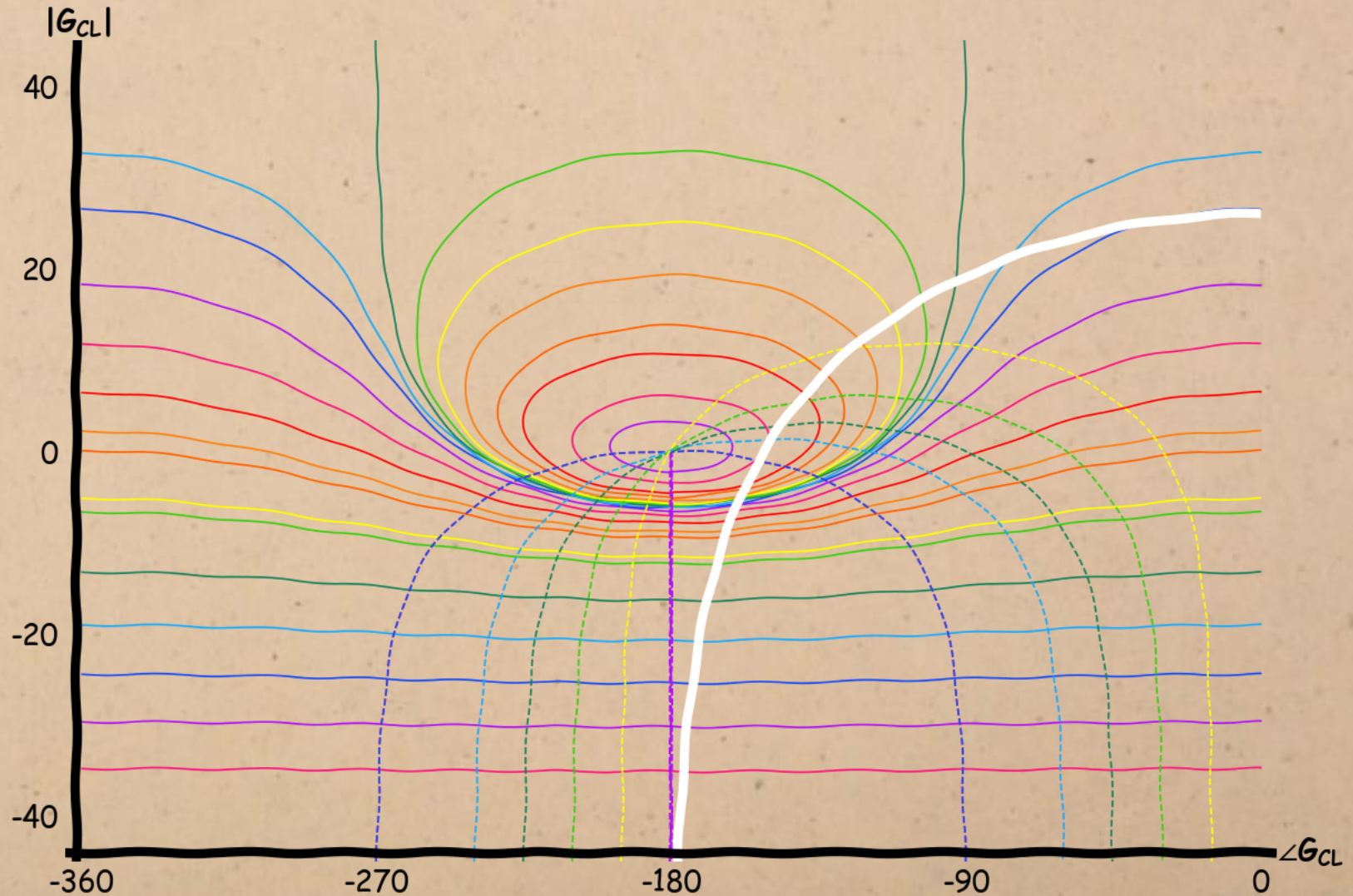
SECOND-ORDER SYSTEM

example 2:



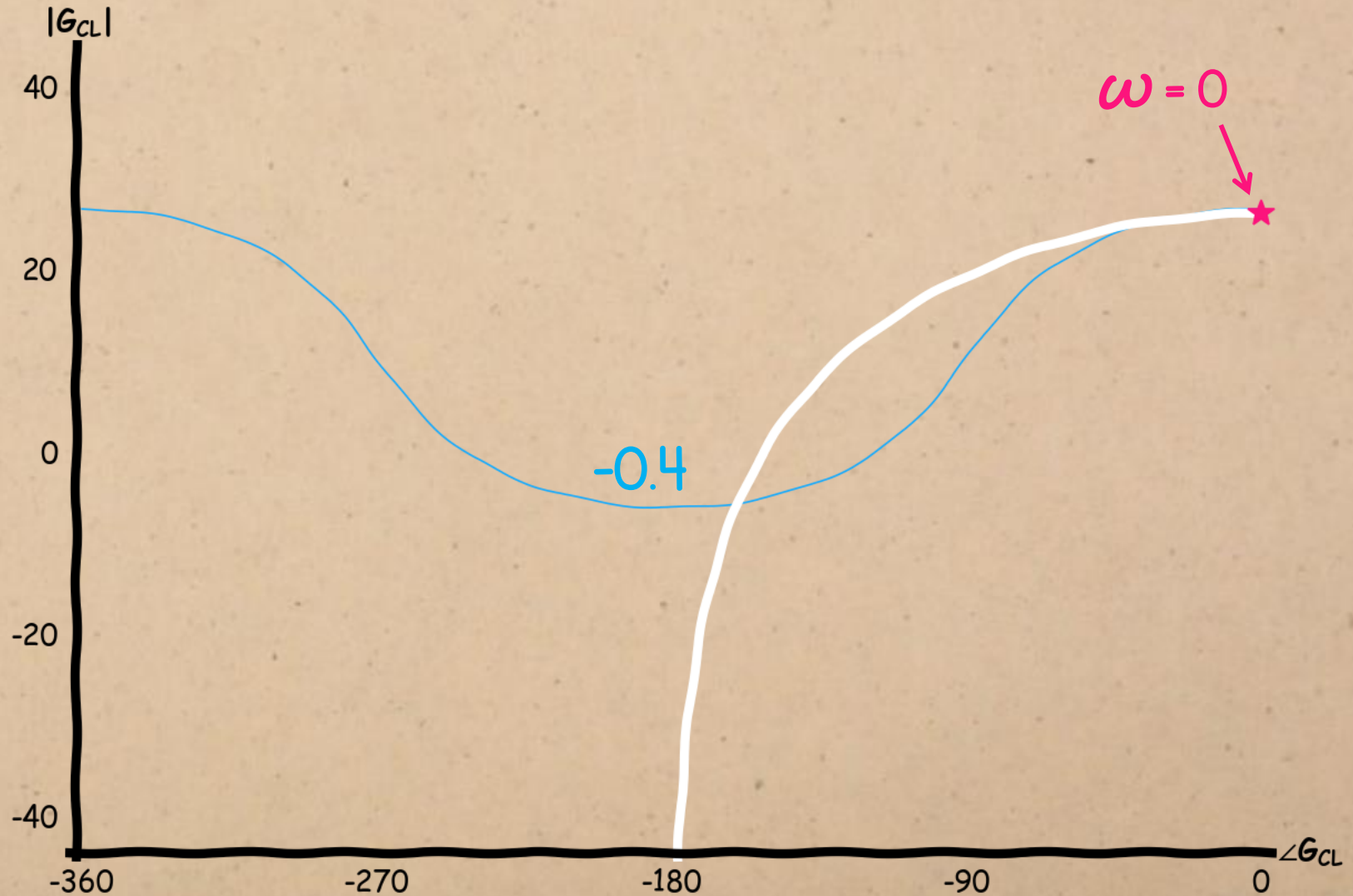
SECOND-ORDER SYSTEM

To the Nichols chart...



SECOND-ORDER SYSTEM

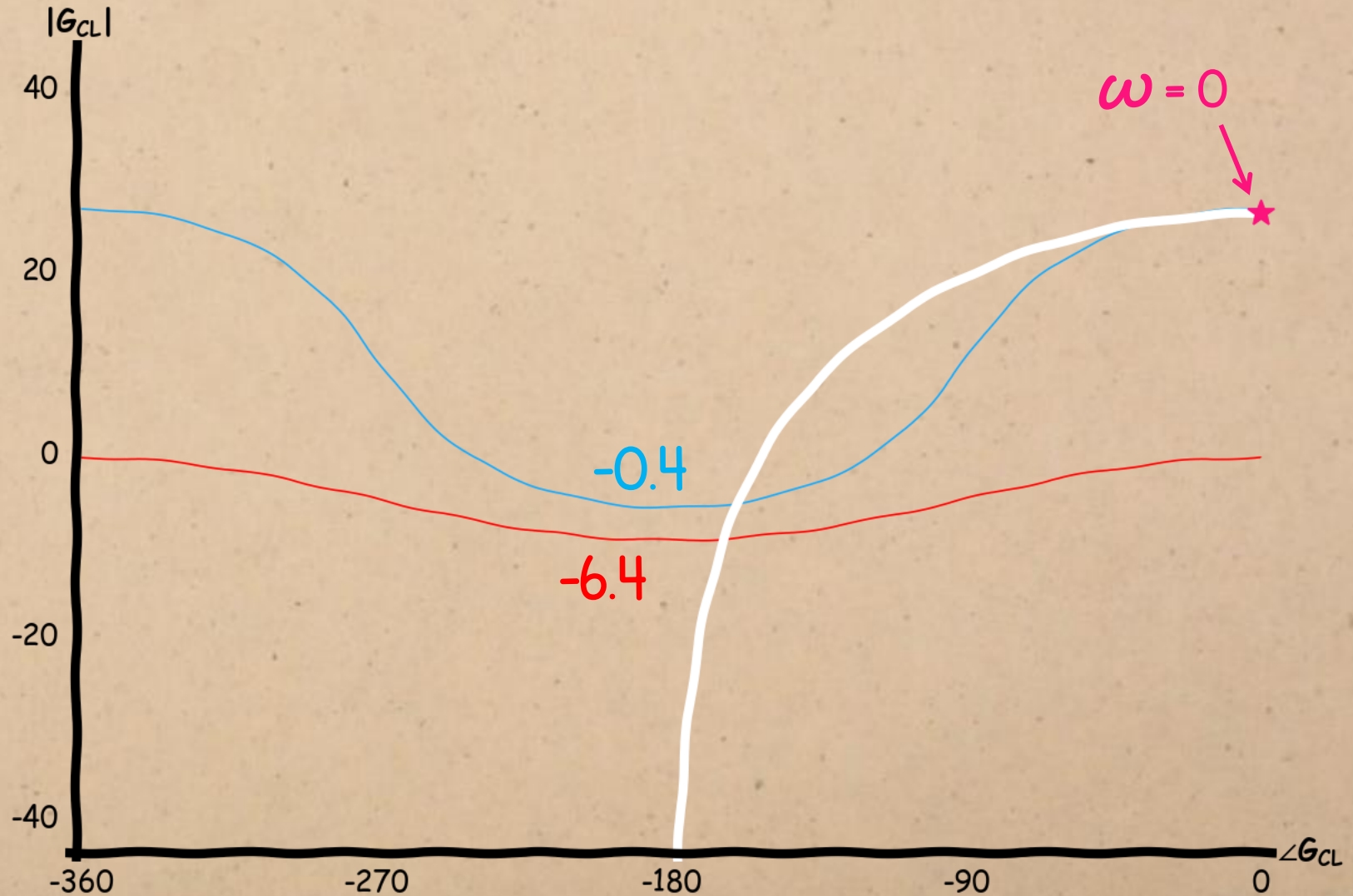
The closed-loop DC gain
is -0.4 dB



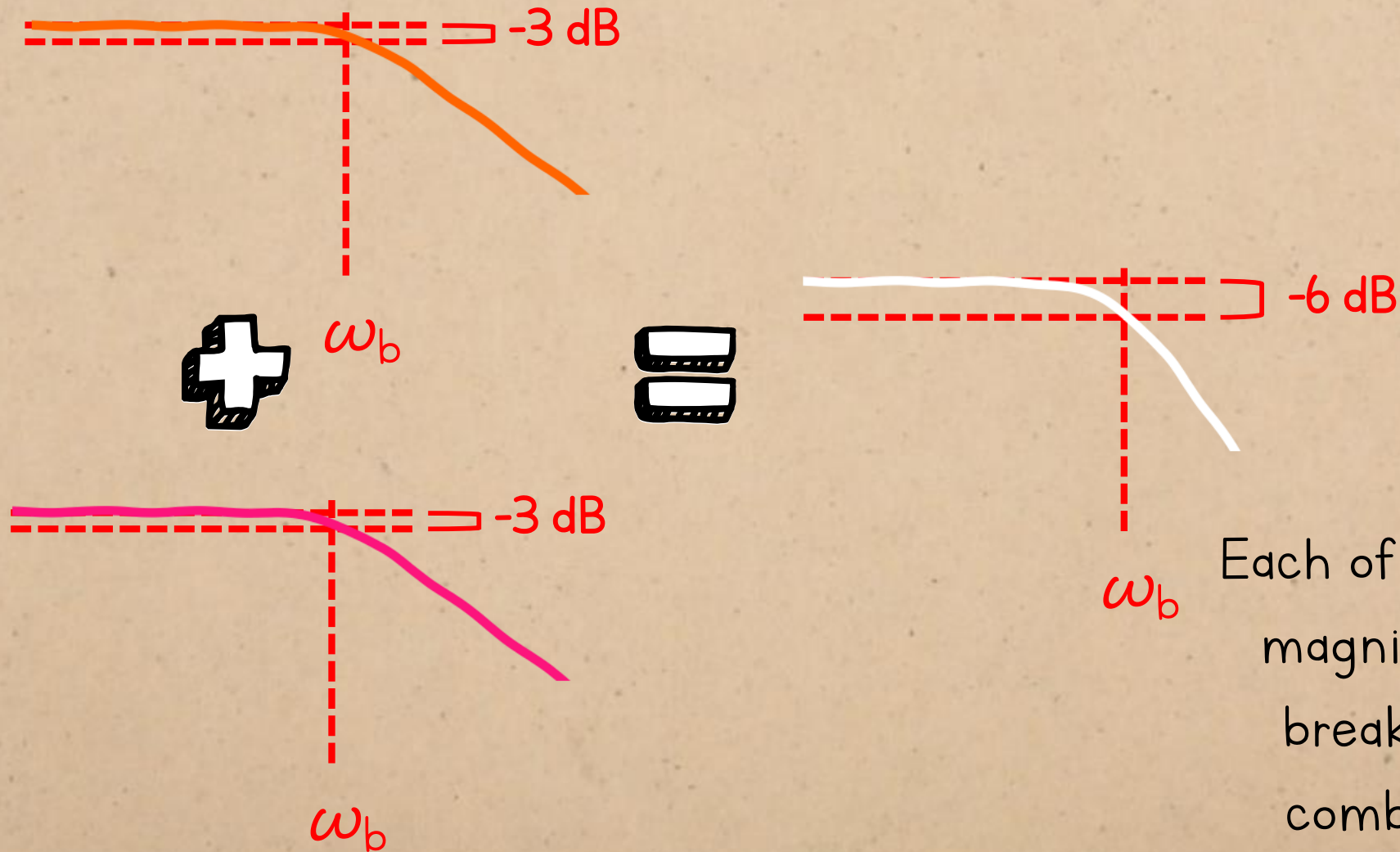
SECOND-ORDER SYSTEM

∴ we need to find where it intersects the -6.4 dB circle

Why not -3.4 dB?



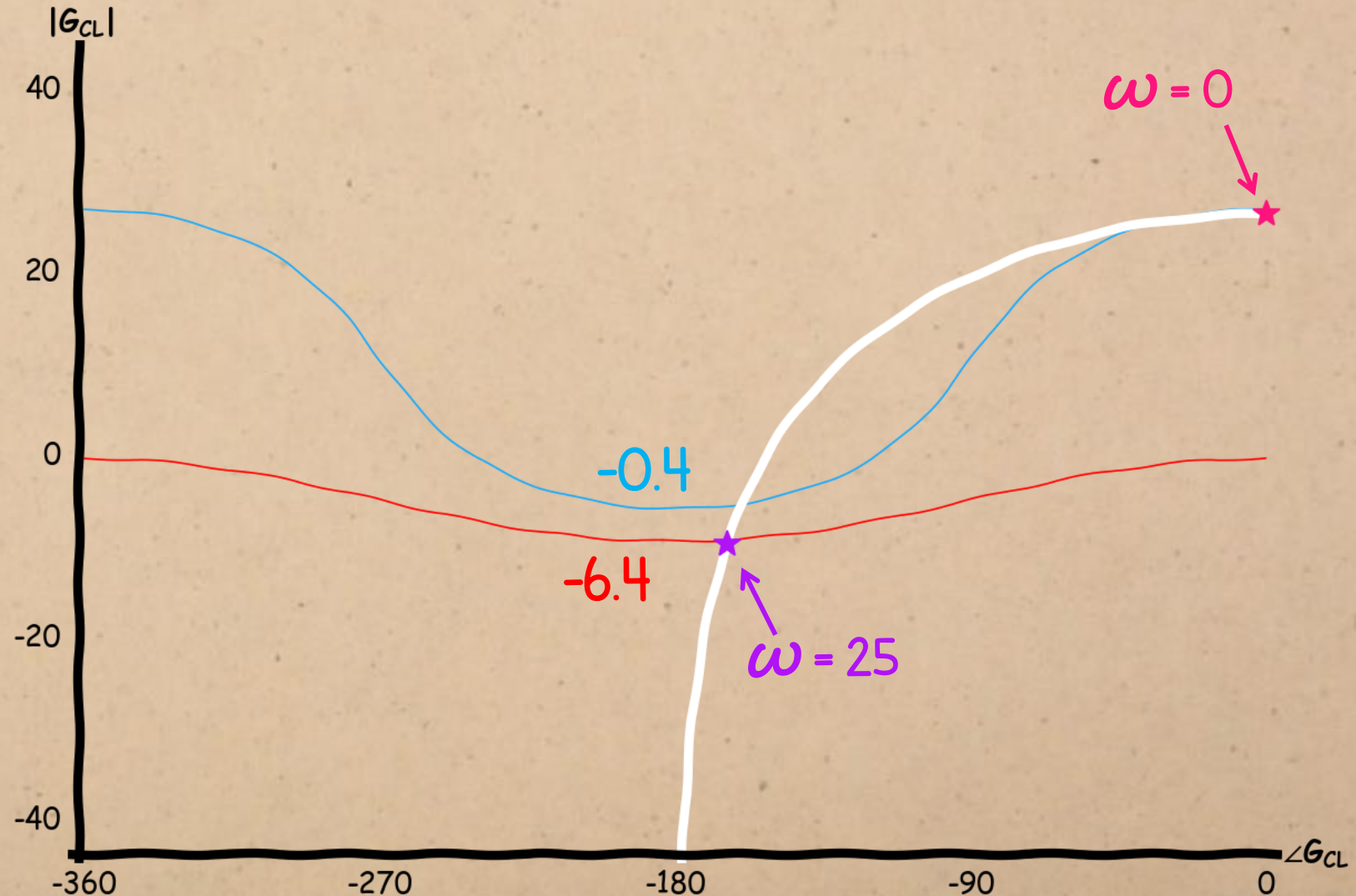
SECOND-ORDER SYSTEM



Each of the **two poles** gives a magnitude of -3 dB at its break frequency, so the combined drop is -6 dB

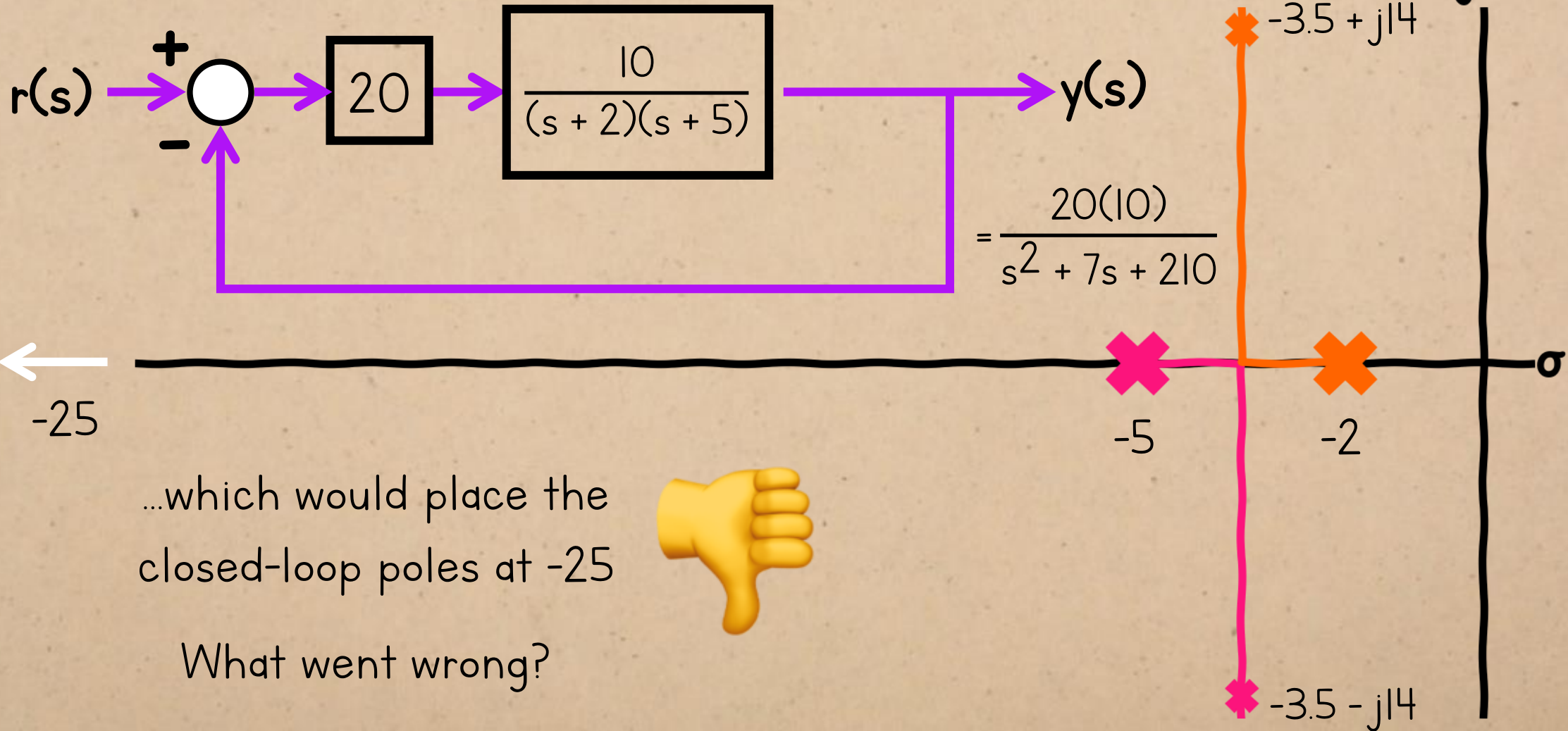
SECOND-ORDER SYSTEM

This approach suggests that the closed-loop break frequency is 25 rad/s

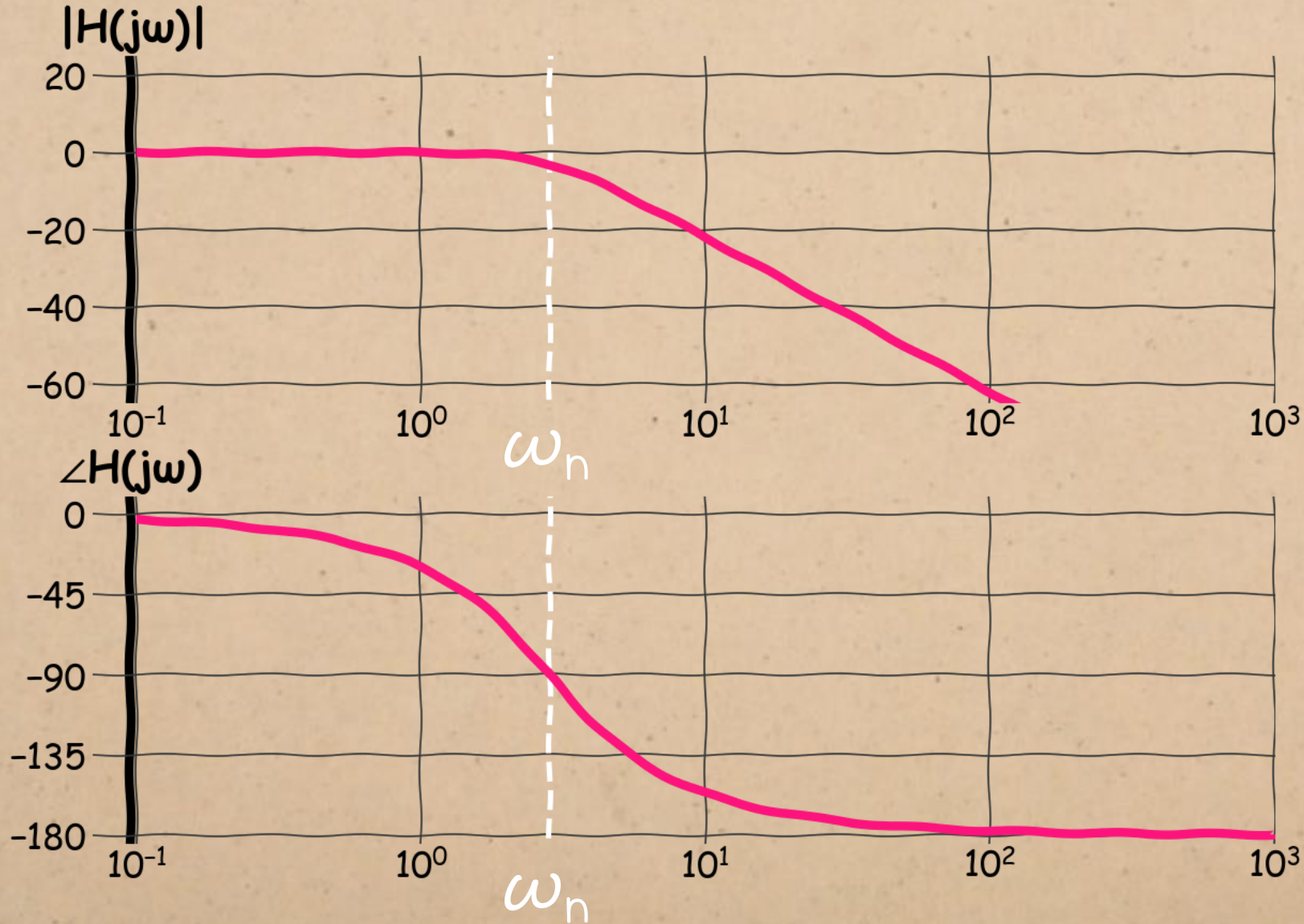


SECOND-ORDER SYSTEM

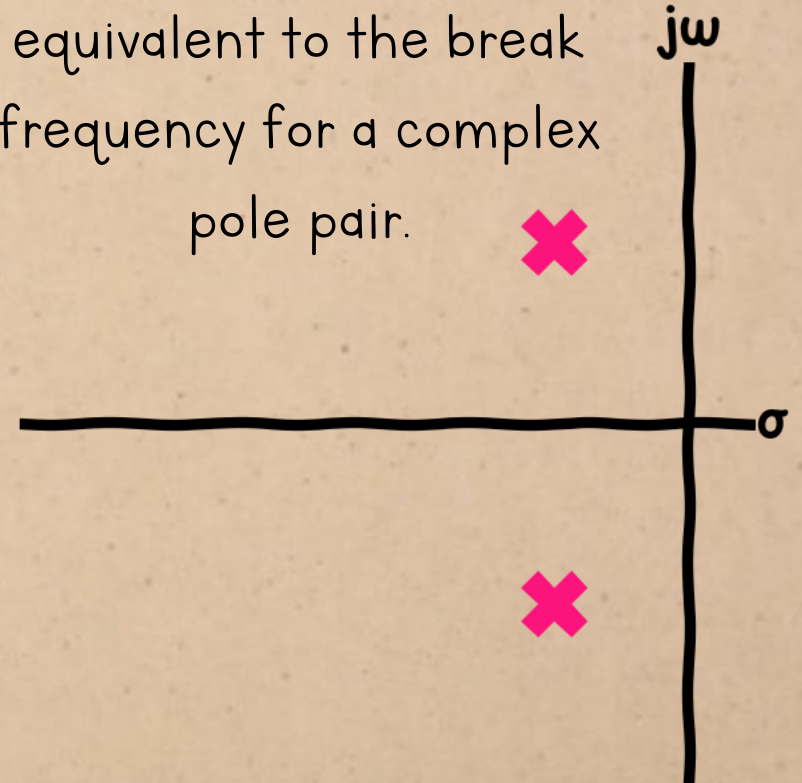
example 2:



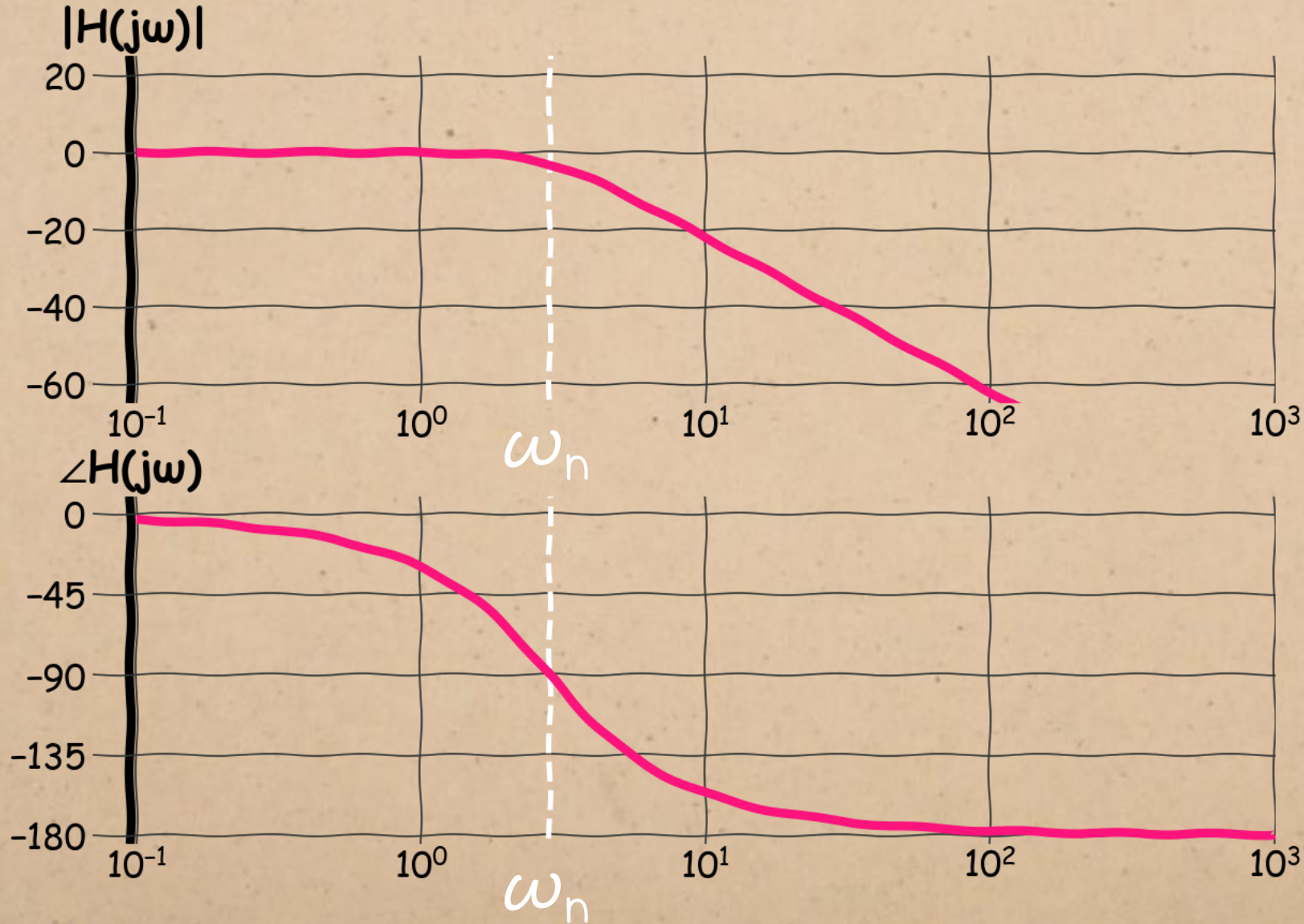
SECOND-ORDER SYSTEM



The natural frequency is equivalent to the break frequency for a complex pole pair.



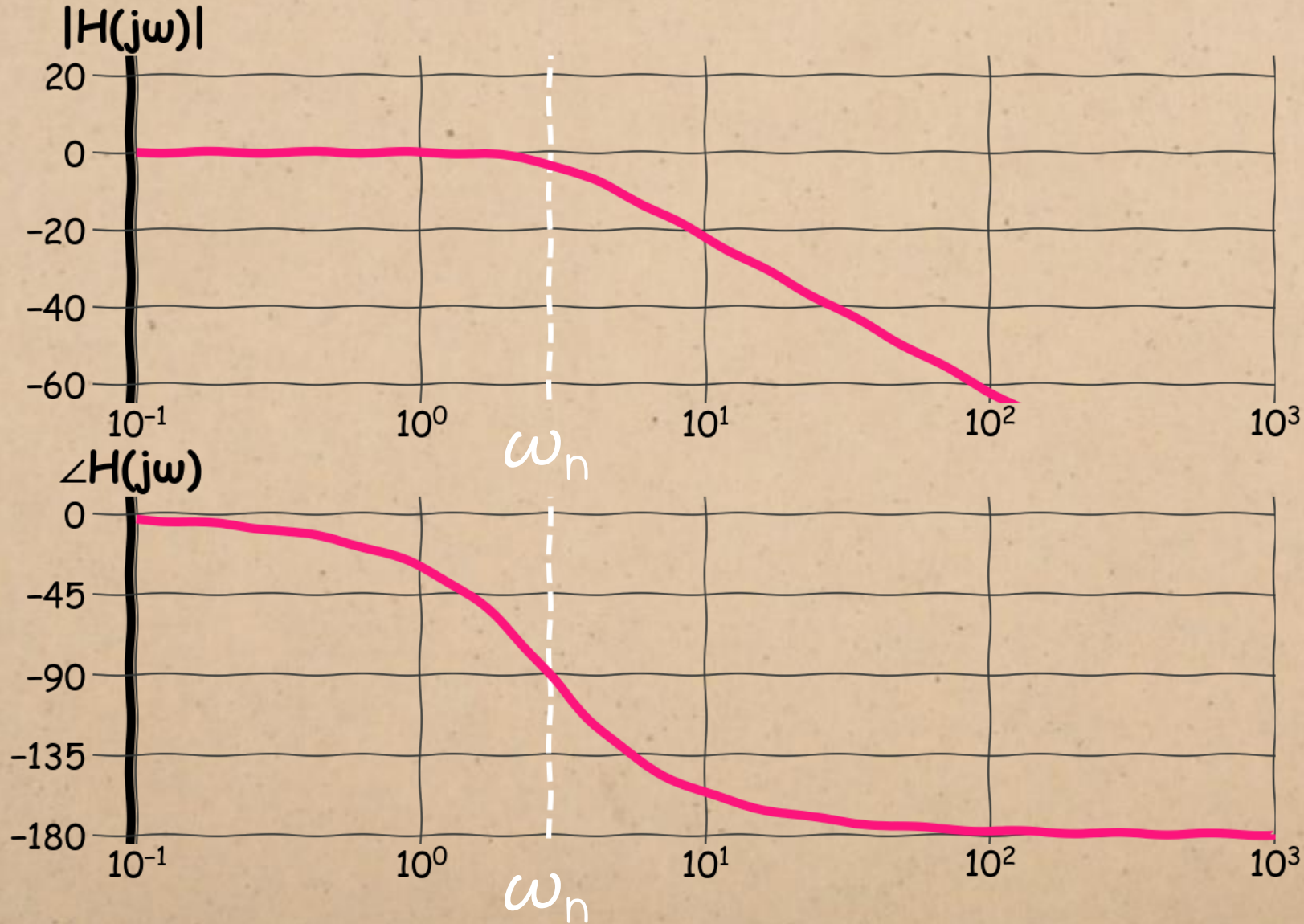
SECOND-ORDER SYSTEM



Finding the general
second-order frequency
response at ω_n ...

$$\frac{A\omega_n^2}{(j\omega_n)^2 + 2\zeta\omega_n(j\omega_n) + \omega_n^2}$$

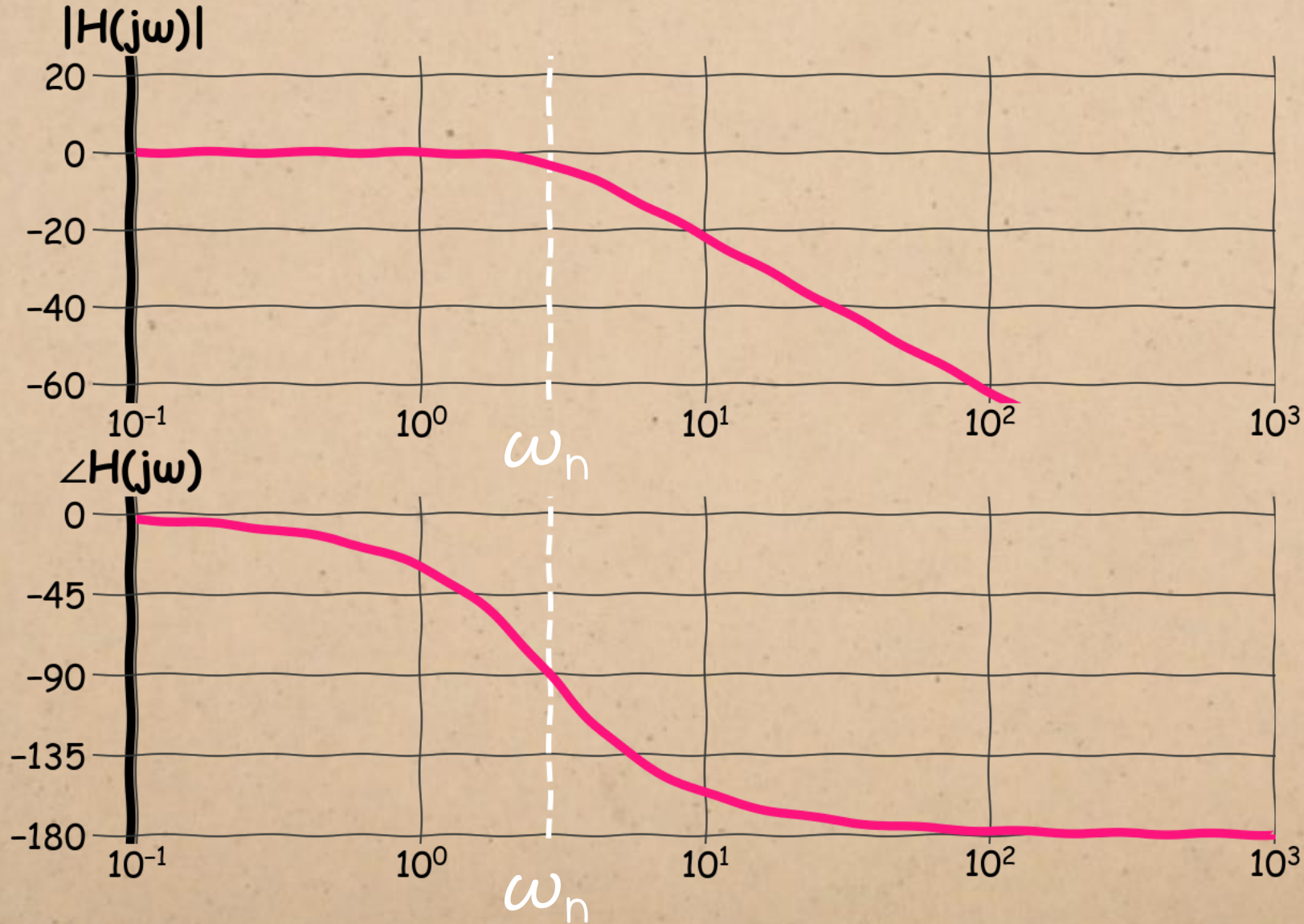
SECOND-ORDER SYSTEM



Finding the general second-order frequency response at ω_n ...

$$\frac{A\omega_n^2}{-\omega_n^2 + j2\zeta\omega_n^2 + \omega_n^2}$$

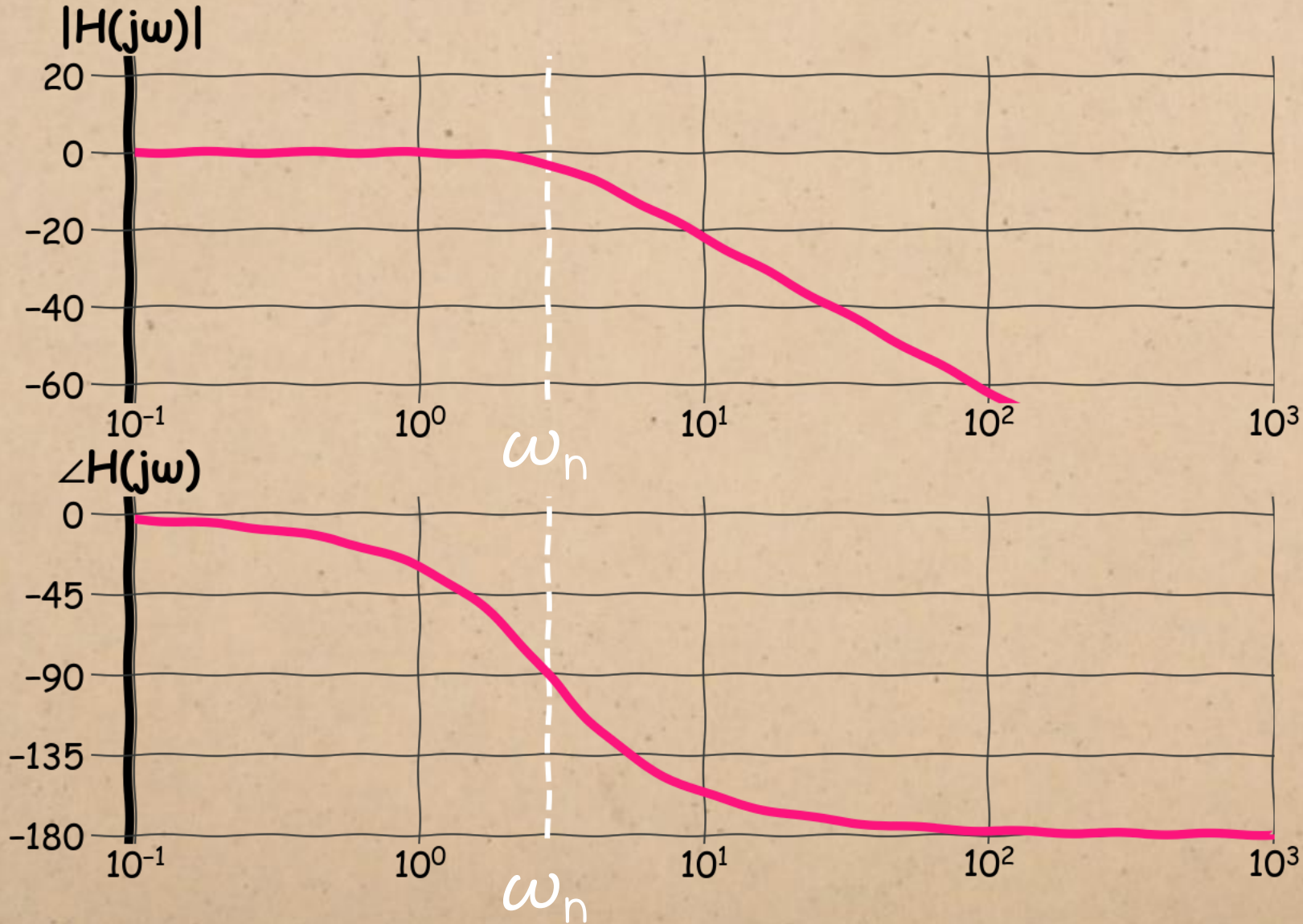
SECOND-ORDER SYSTEM



Finding the general
second-order frequency
response at ω_n ...

$$\frac{A\omega_n^2}{j2\zeta\omega_n^2}$$

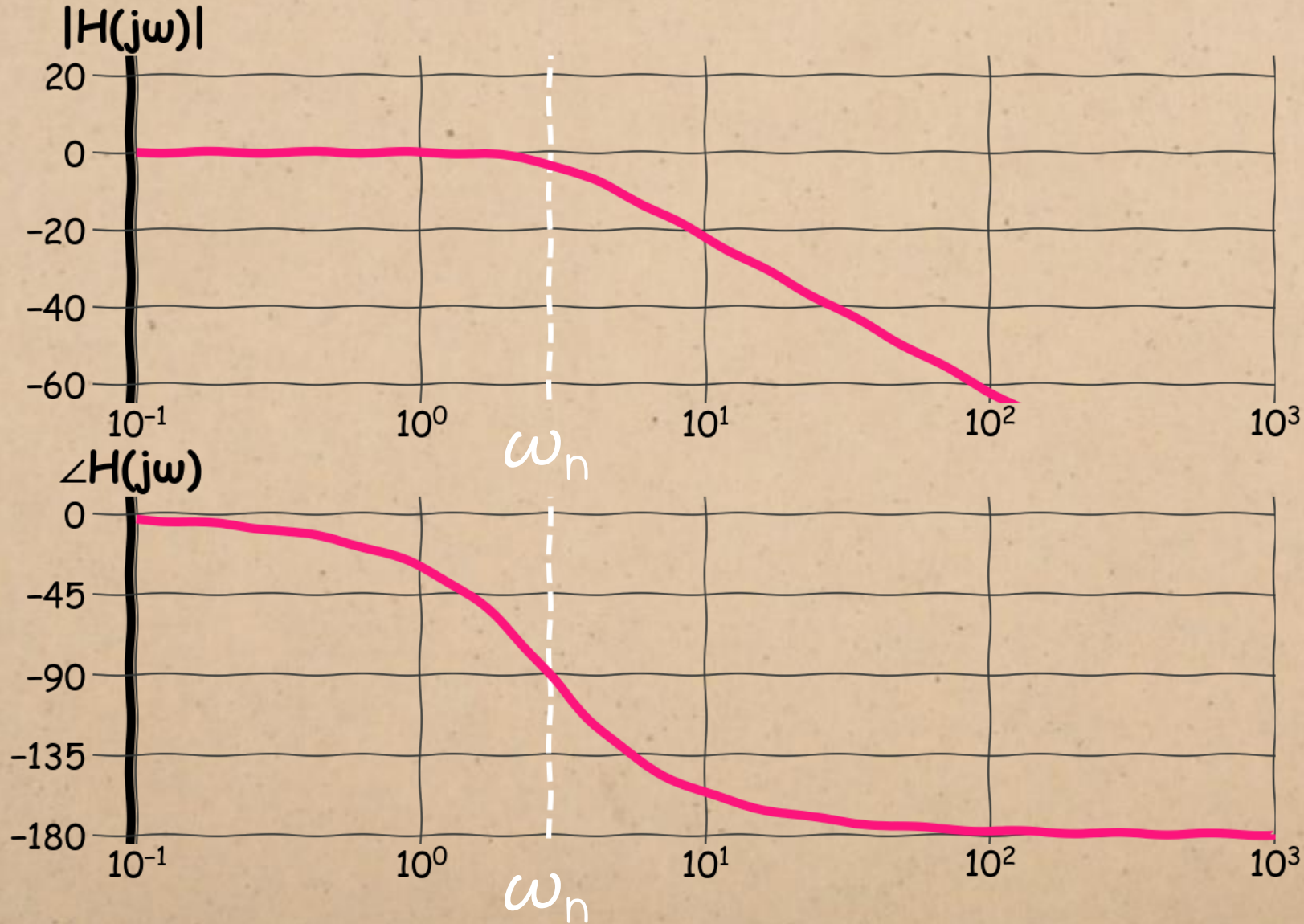
SECOND-ORDER SYSTEM



Finding the general
second-order frequency
response at ω_n ...

$$\frac{A}{j2\zeta}$$

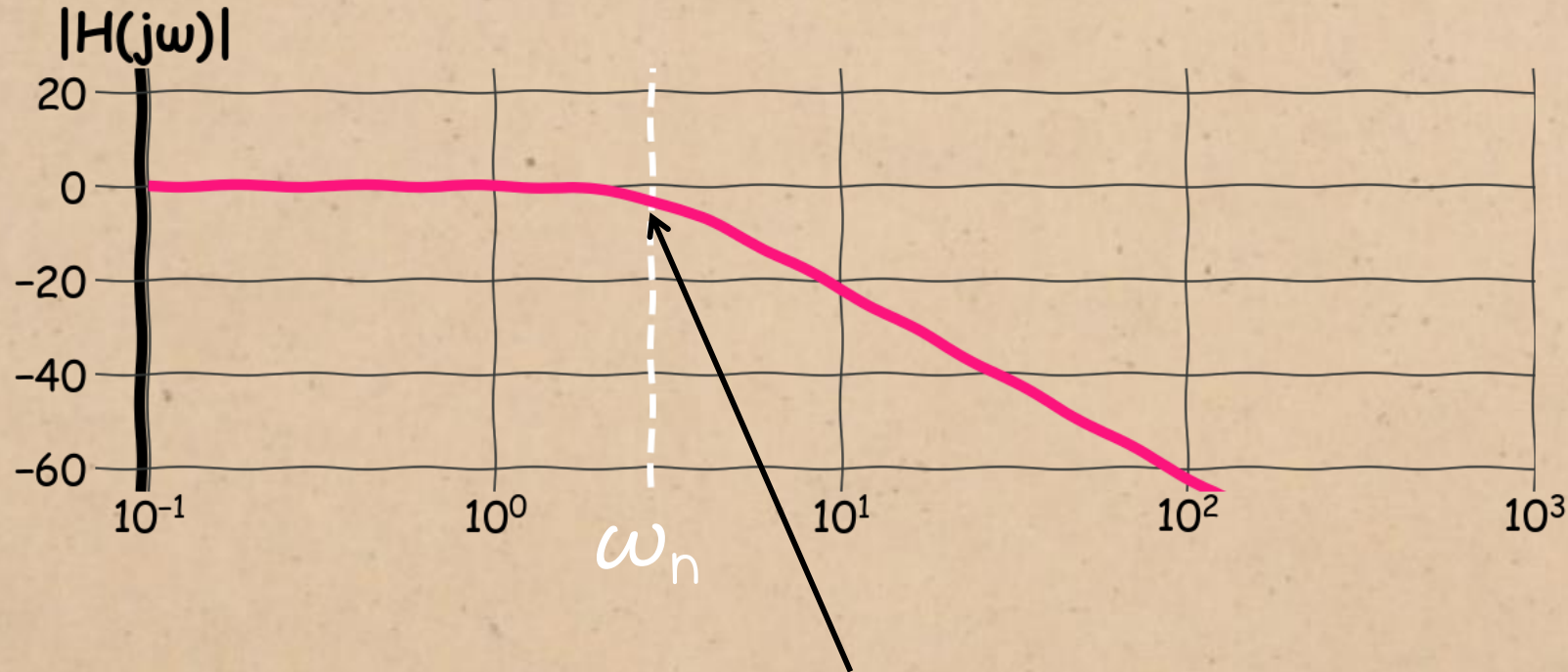
SECOND-ORDER SYSTEM



$\therefore$ The magnitude at ω_n :

$$|G(j\omega_n)| = \left| \frac{A}{j2\zeta} \right| = \frac{A}{2\zeta}$$

SECOND-ORDER SYSTEM

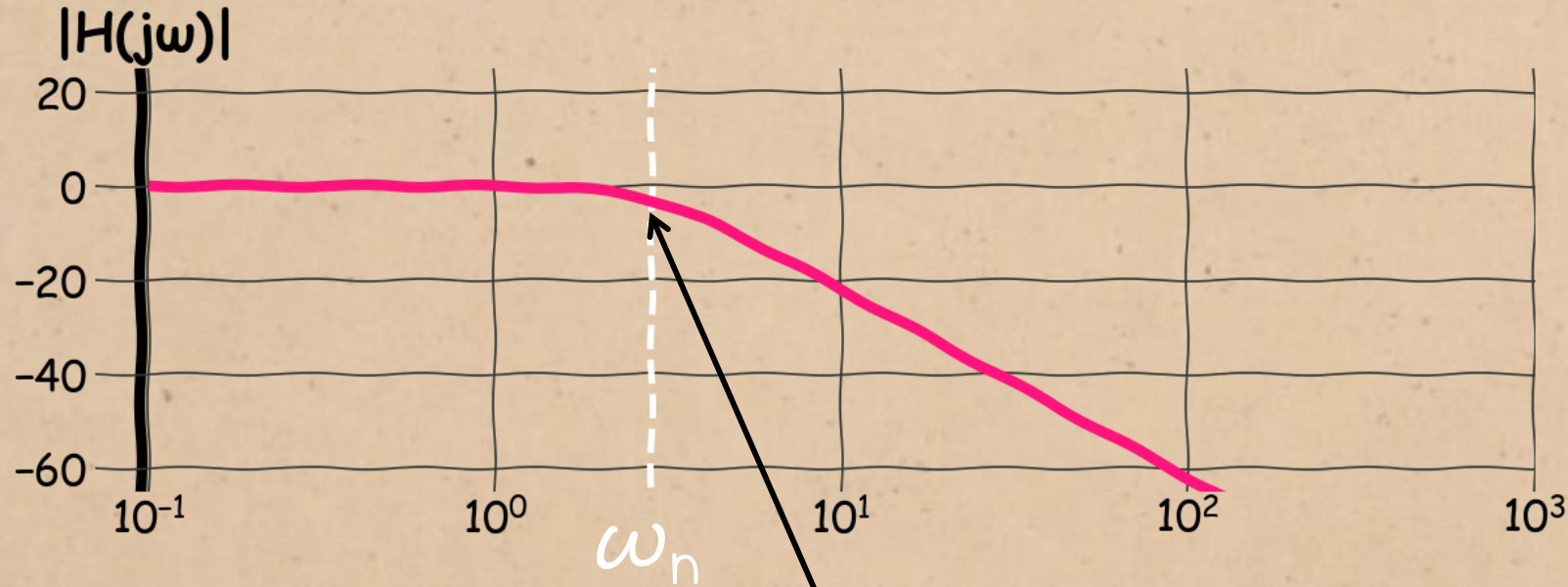


$$= 20 \log_{10} A - 20 \log_{10} 2 - 20 \log_{10} \zeta$$

$\therefore$ The magnitude at ω_n :

$$|G(j\omega_n)| = \left| \frac{A}{j2\zeta} \right| = \frac{A}{2\zeta}$$

SECOND-ORDER SYSTEM



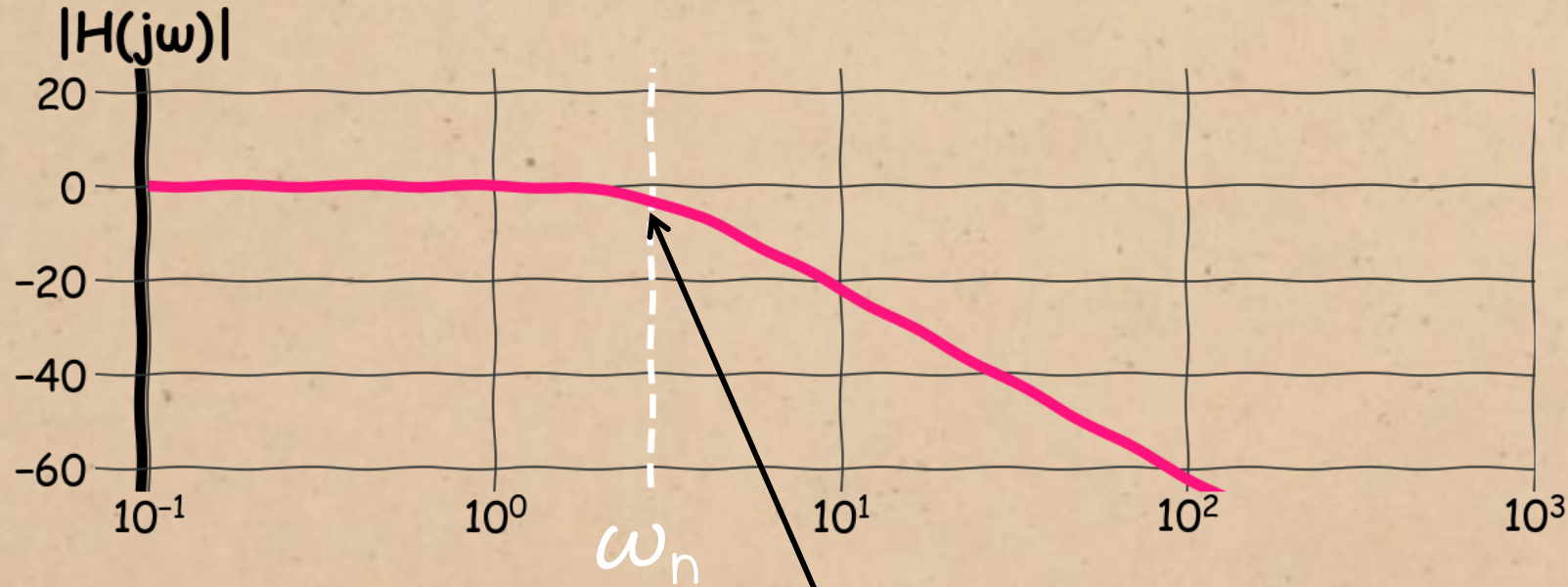
$$= 20 \log_{10} A - 6 - 20 \log_{10} \zeta$$

∴ The magnitude at ω_n :

$$|G(j\omega_n)| = \left| \frac{A}{j2\zeta} \right| = \frac{A}{2\zeta}$$

So we still get the DC gain - 6 dB, as expected...

SECOND-ORDER SYSTEM



$$= 20 \log_{10} A - 6 - 20 \log_{10} \zeta$$

but we also have to take the damping ratio into account.

∴ The magnitude at ω_n :

$$|G(j\omega_n)| = \left| \frac{A}{j2\zeta} \right| = \frac{A}{2} \frac{1}{\zeta}$$

SECOND-ORDER SYSTEM

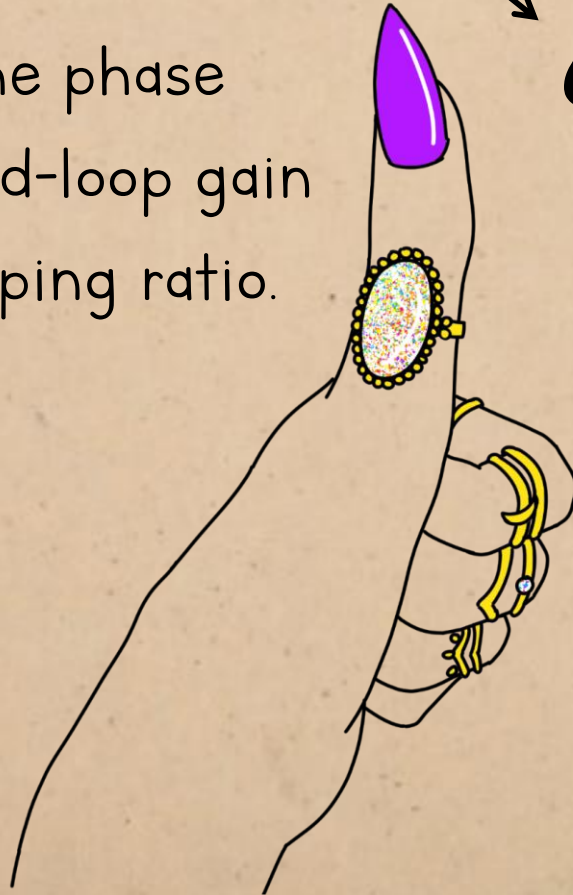
damping ratio

closed-loop gain

We can use the phase margin and closed-loop gain to find the damping ratio.

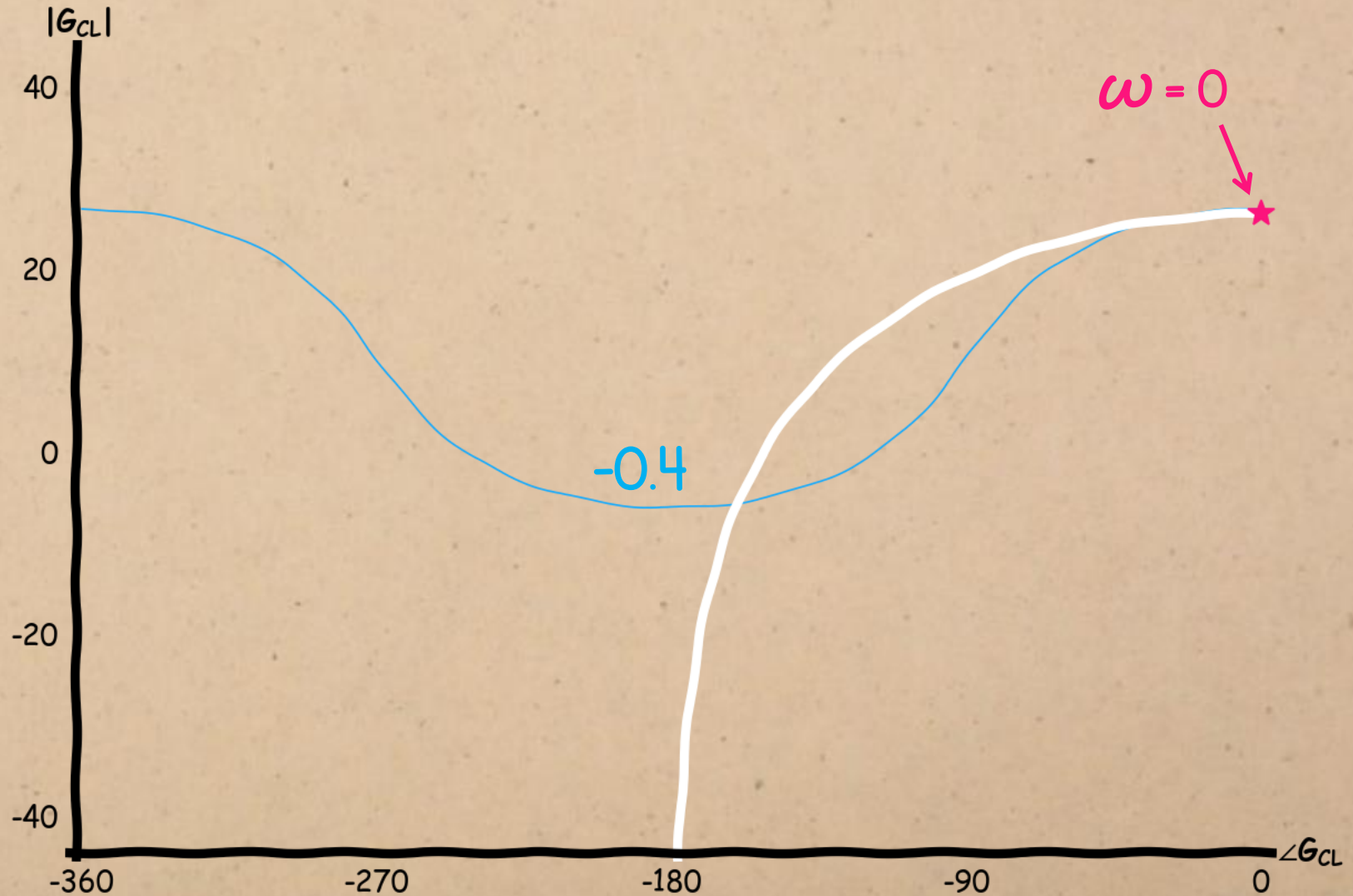
$$\zeta = \frac{A}{100} \phi_m$$

phase margin (in degrees)



SECOND-ORDER SYSTEM

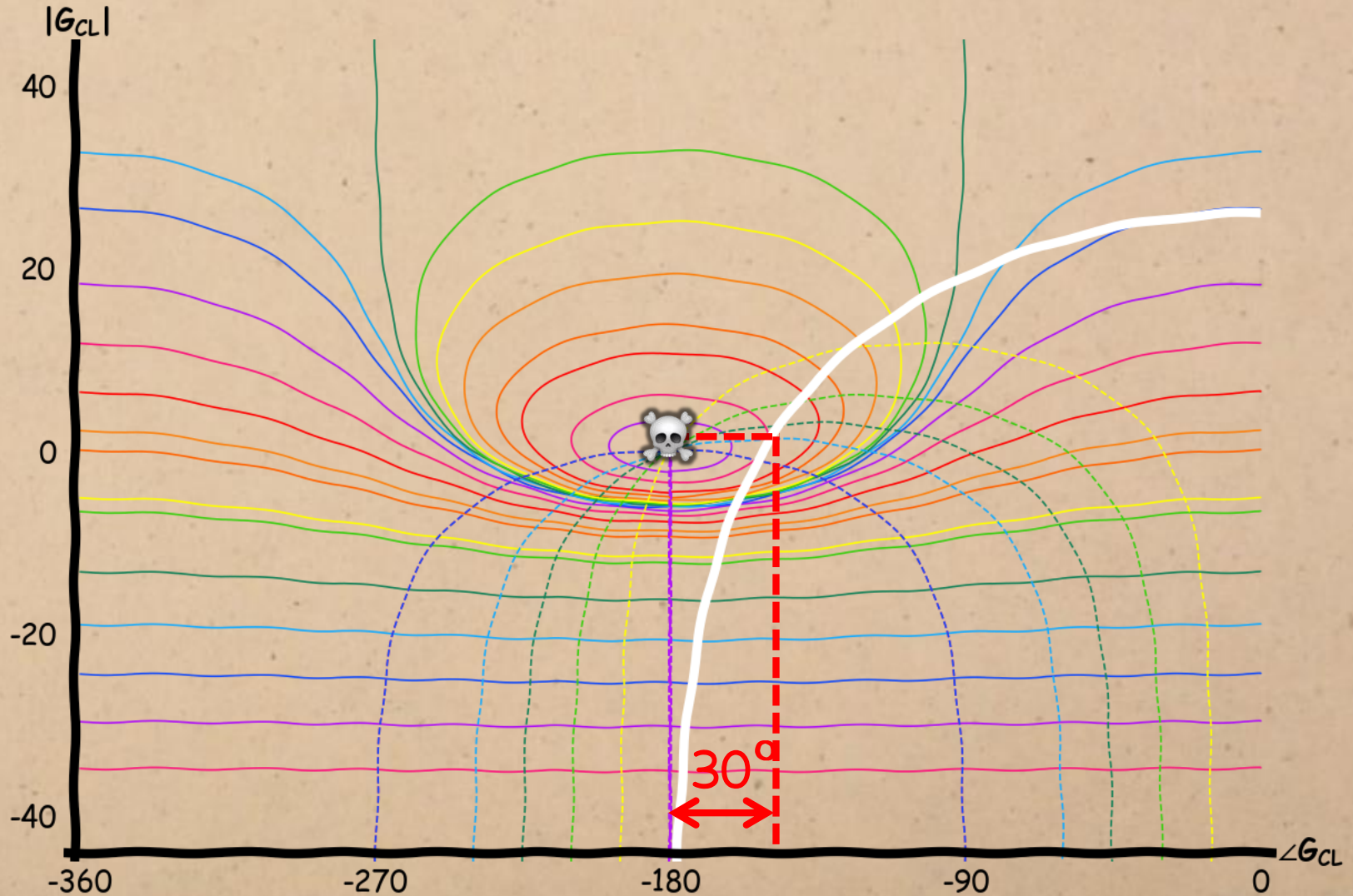
We know the closed-loop
DC gain is -0.4 dB
 $= 0.95$



SECOND-ORDER SYSTEM

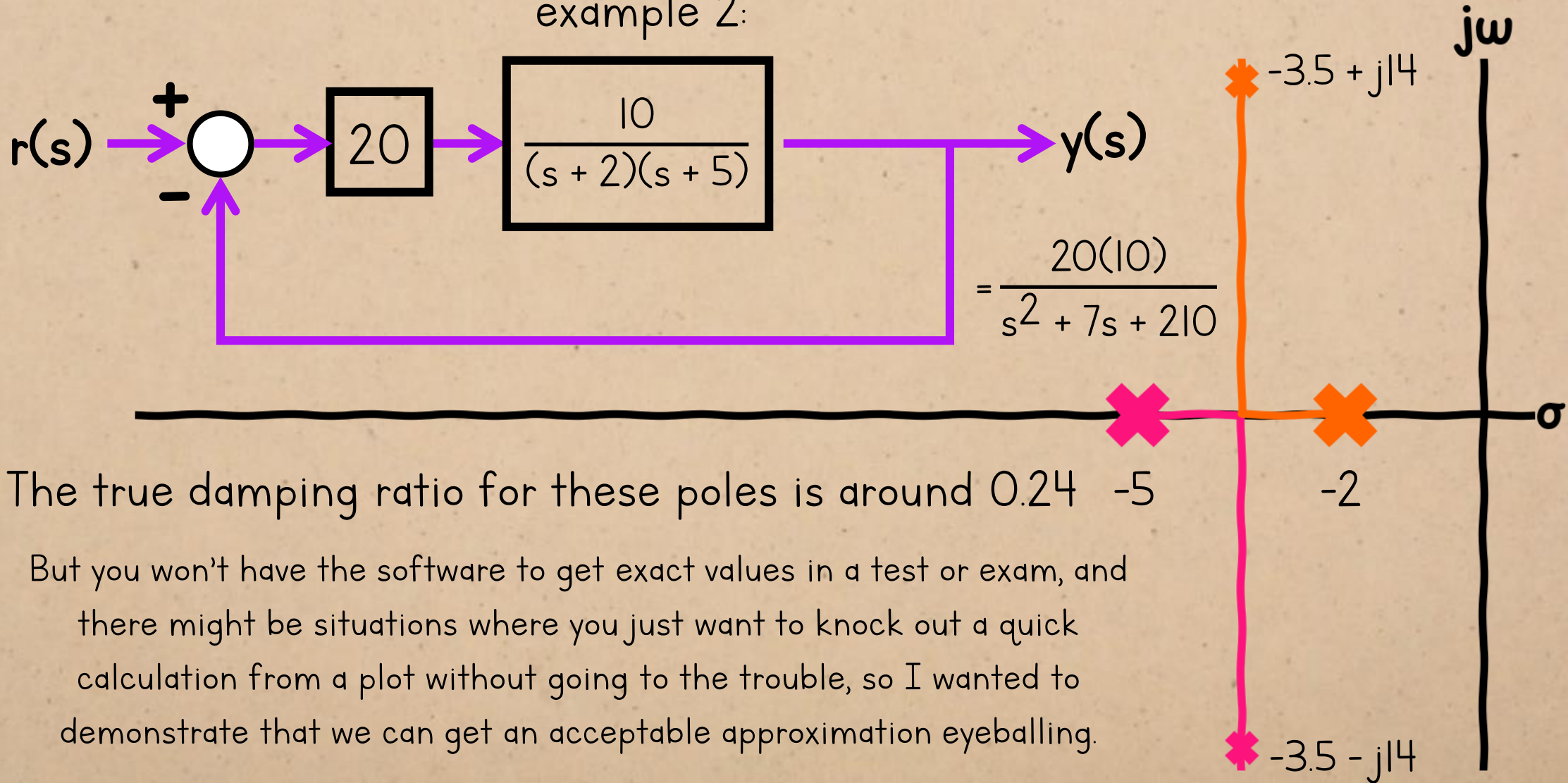
And we can see that the phase margin is around 30°

$$\therefore \zeta = \frac{0.95}{100} 30 = 0.285$$



SECOND-ORDER SYSTEM

example 2:

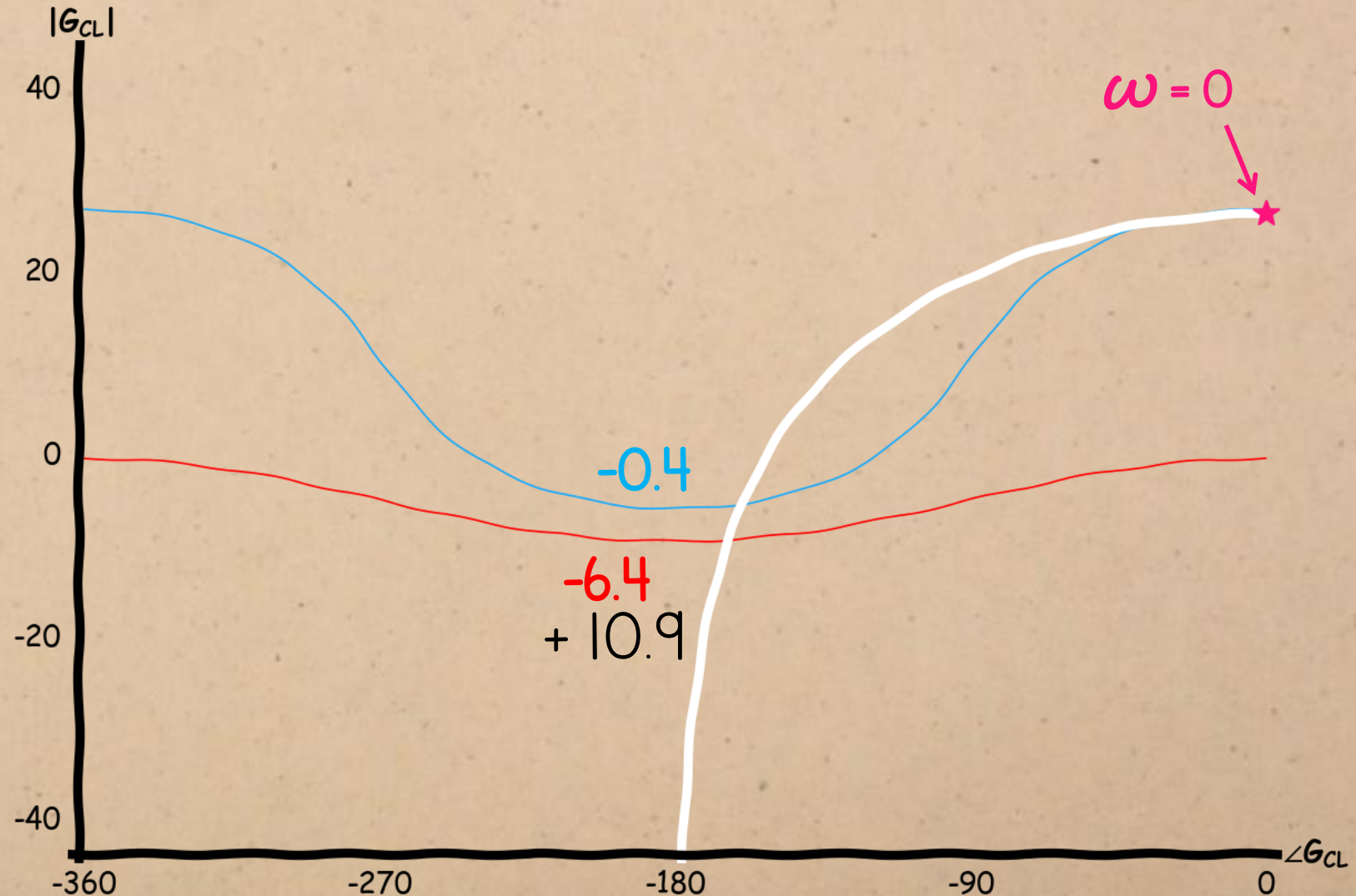


The true damping ratio for these poles is around 0.24

But you won't have the software to get exact values in a test or exam, and there might be situations where you just want to knock out a quick calculation from a plot without going to the trouble, so I wanted to demonstrate that we can get an acceptable approximation eyeballing.

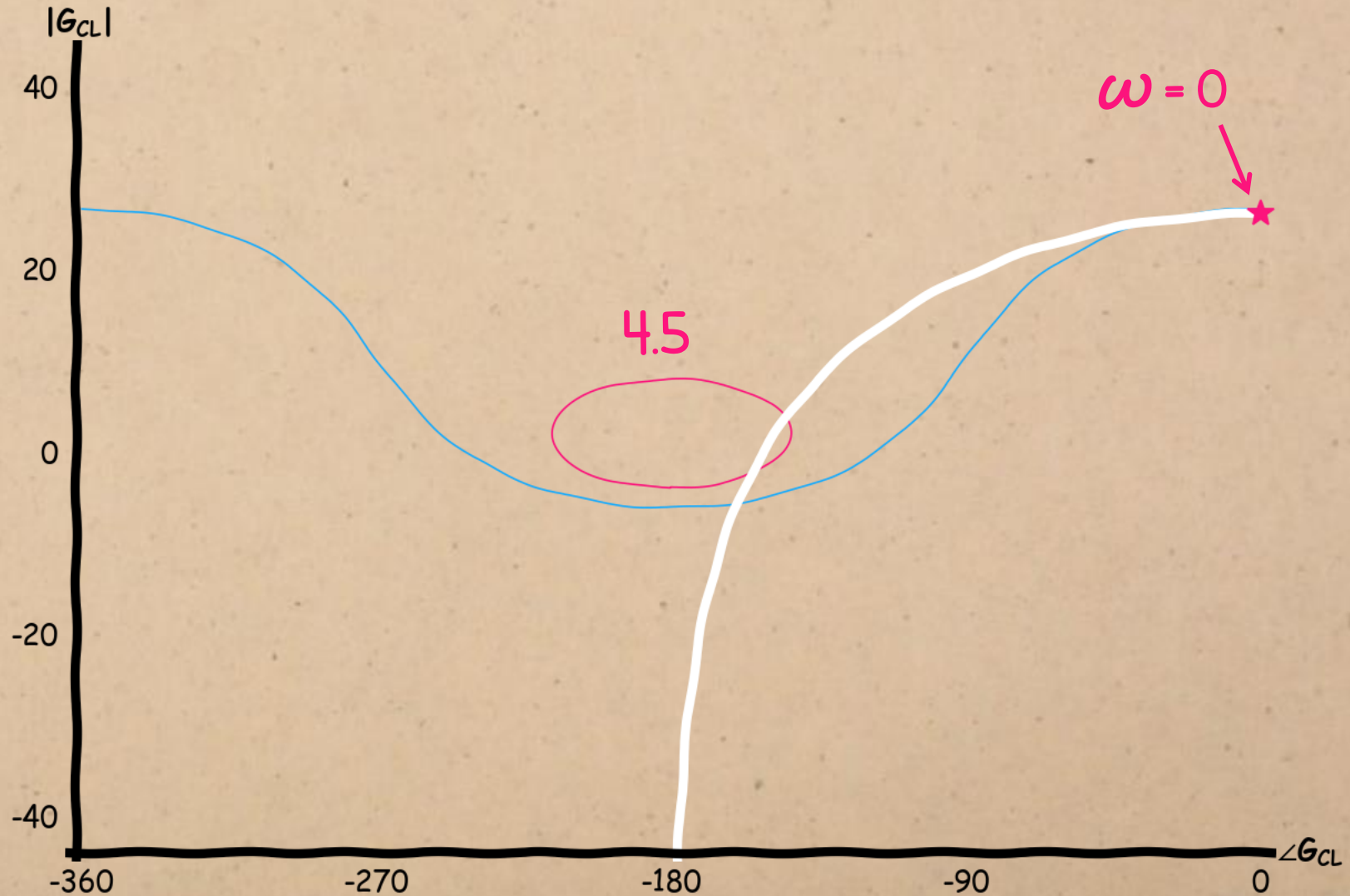
SECOND-ORDER SYSTEM

∴ we need to subtract
 $20\log_{10} 0.285 = -10.9$
from the gain we're
looking for.

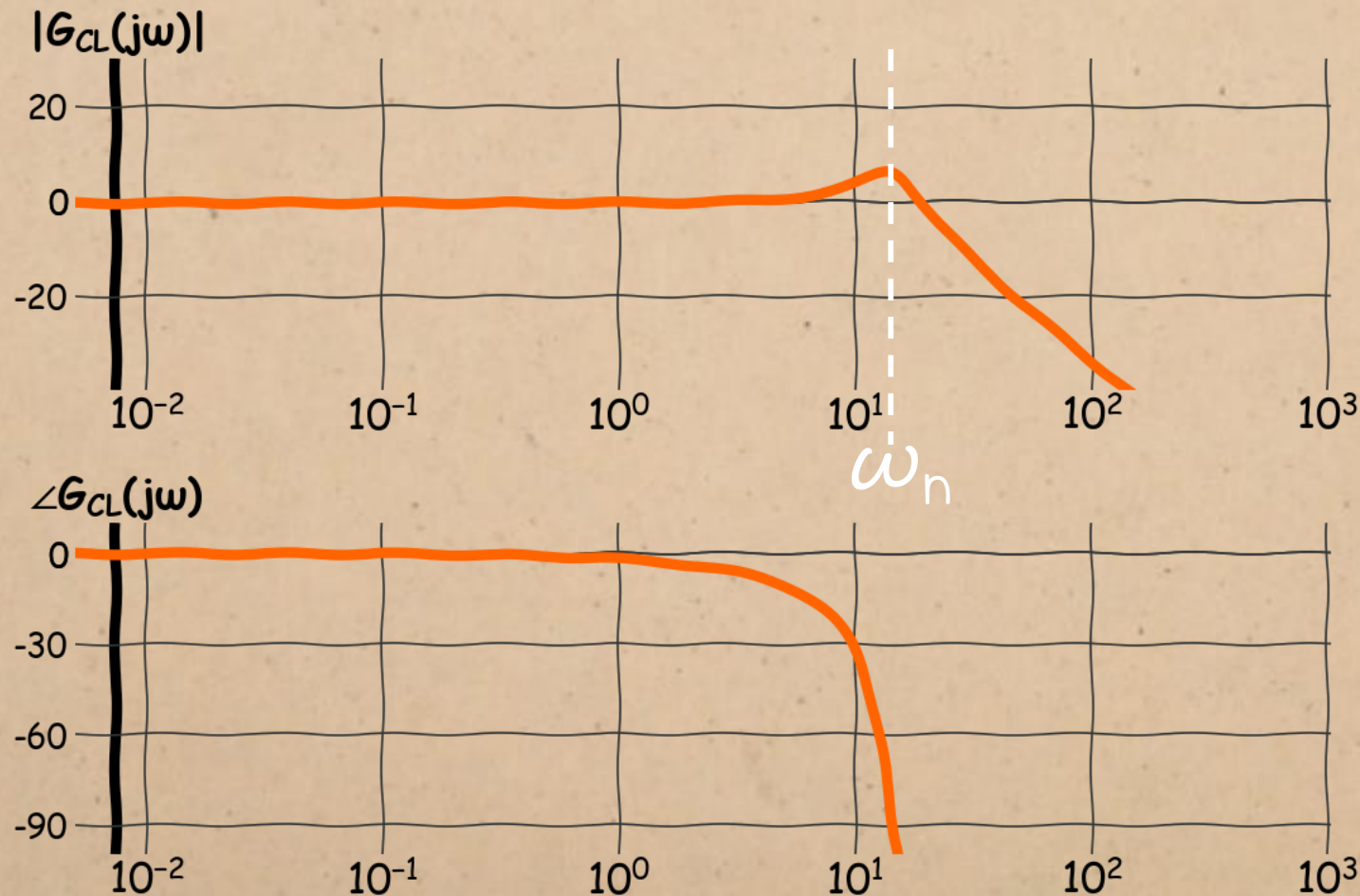


SECOND-ORDER SYSTEM

So now, we're looking for the frequency that intersects the 4.5 dB circle.



SECOND-ORDER SYSTEM

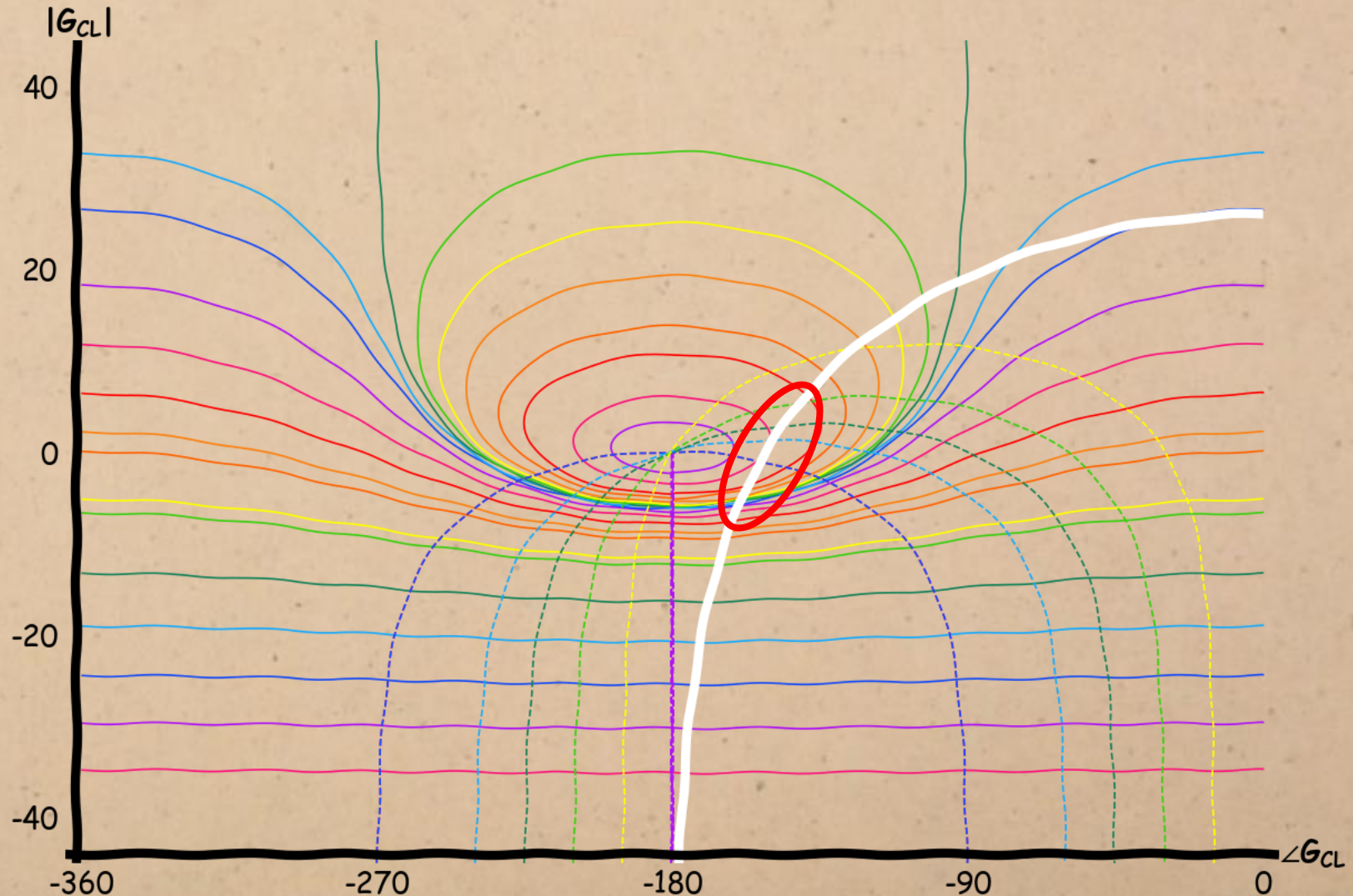


If it seems strange that this value is larger than the DC gain, remember that the closed-loop response will be **resonant** at the natural frequency, because the **damping ratio is small**.

SECOND-ORDER SYSTEM

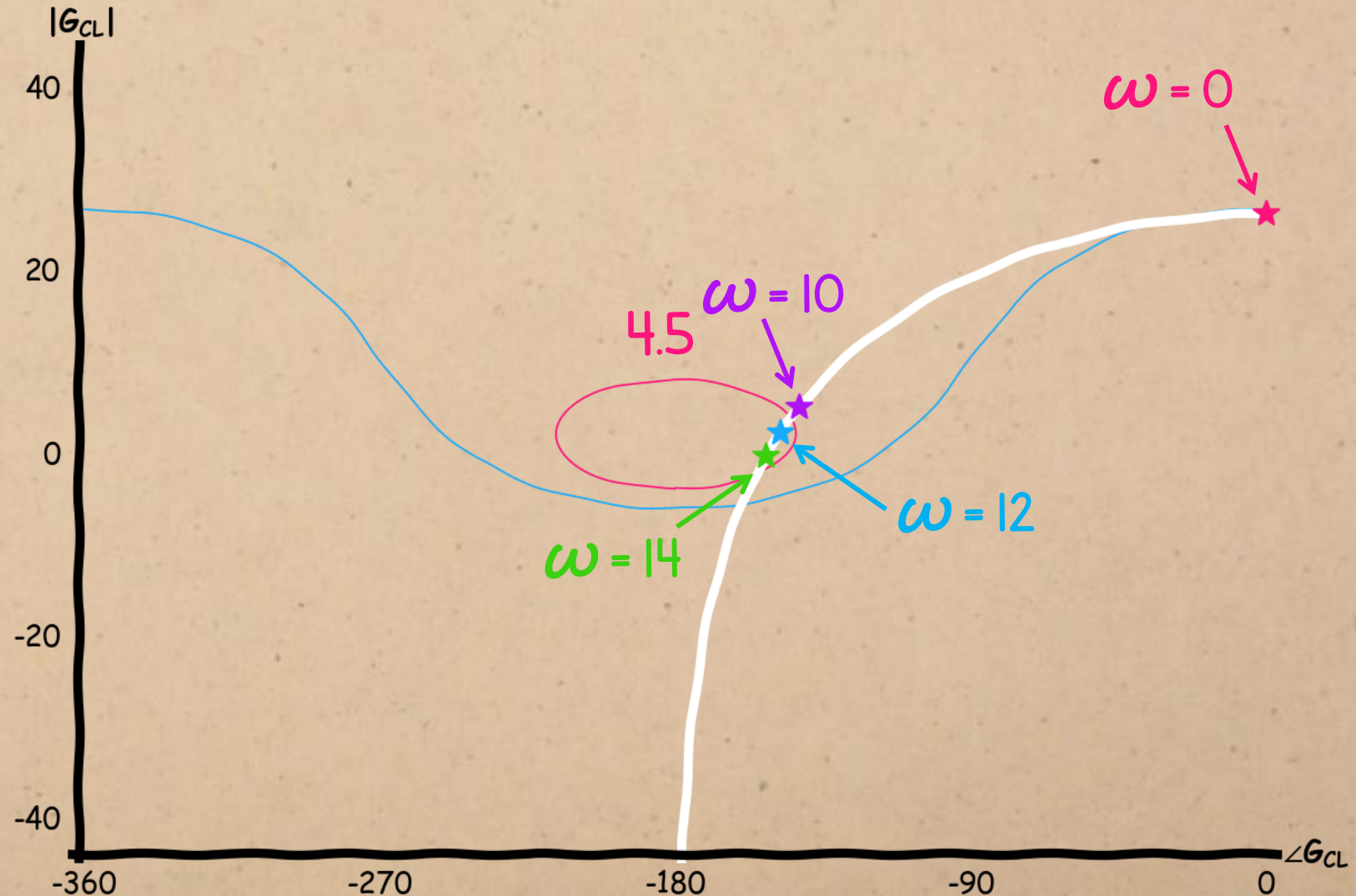
For a resonant system, the **maximum gain** will be at the natural frequency.

So you can just find the **largest M circle** that the response intersects.



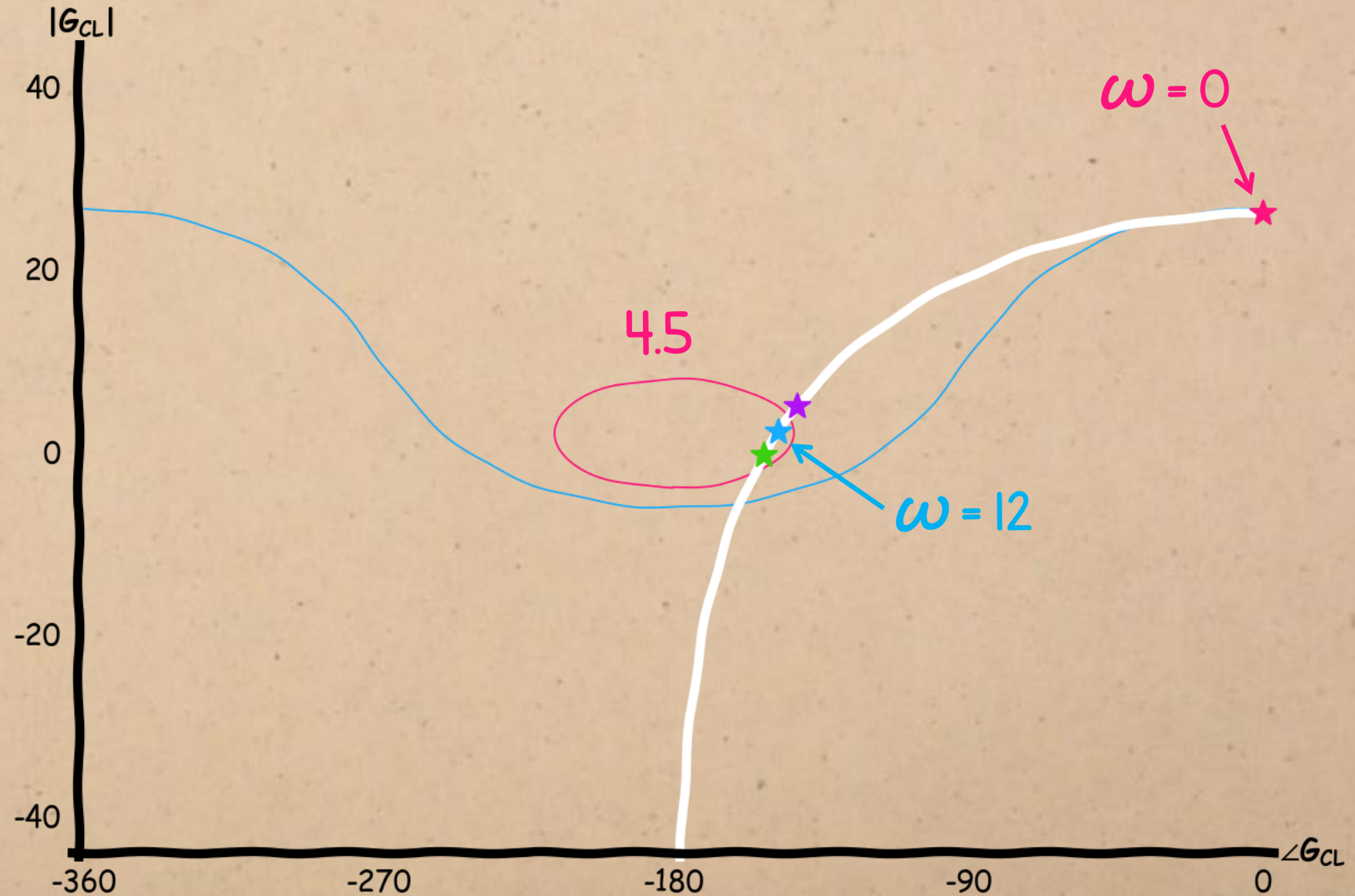
SECOND-ORDER SYSTEM

The intersection isn't perfect because the approximate phase margin we read wasn't perfect, but if we look at the frequencies in this general vicinity...



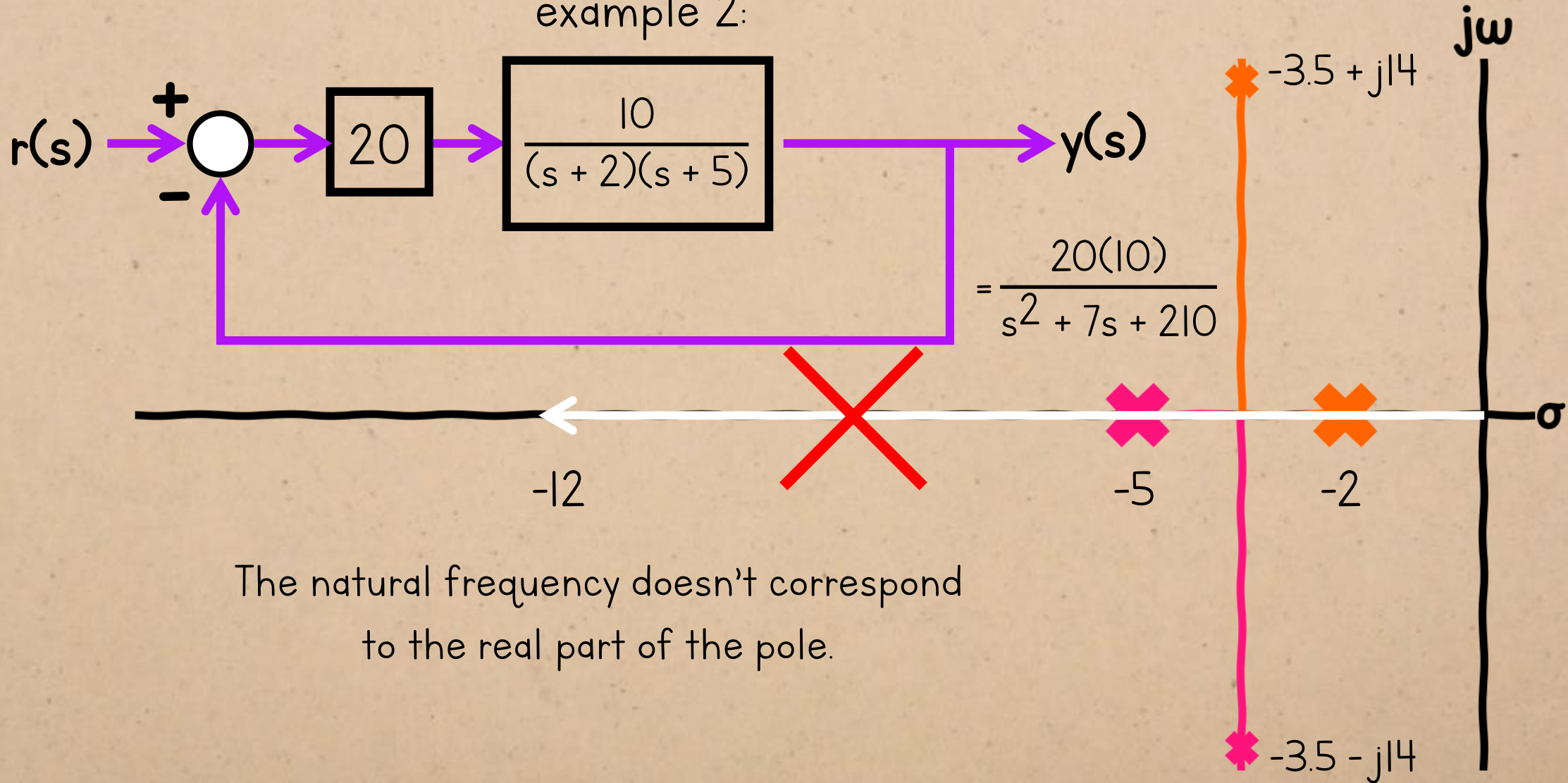
SECOND-ORDER SYSTEM

12 rad/s looks about right.



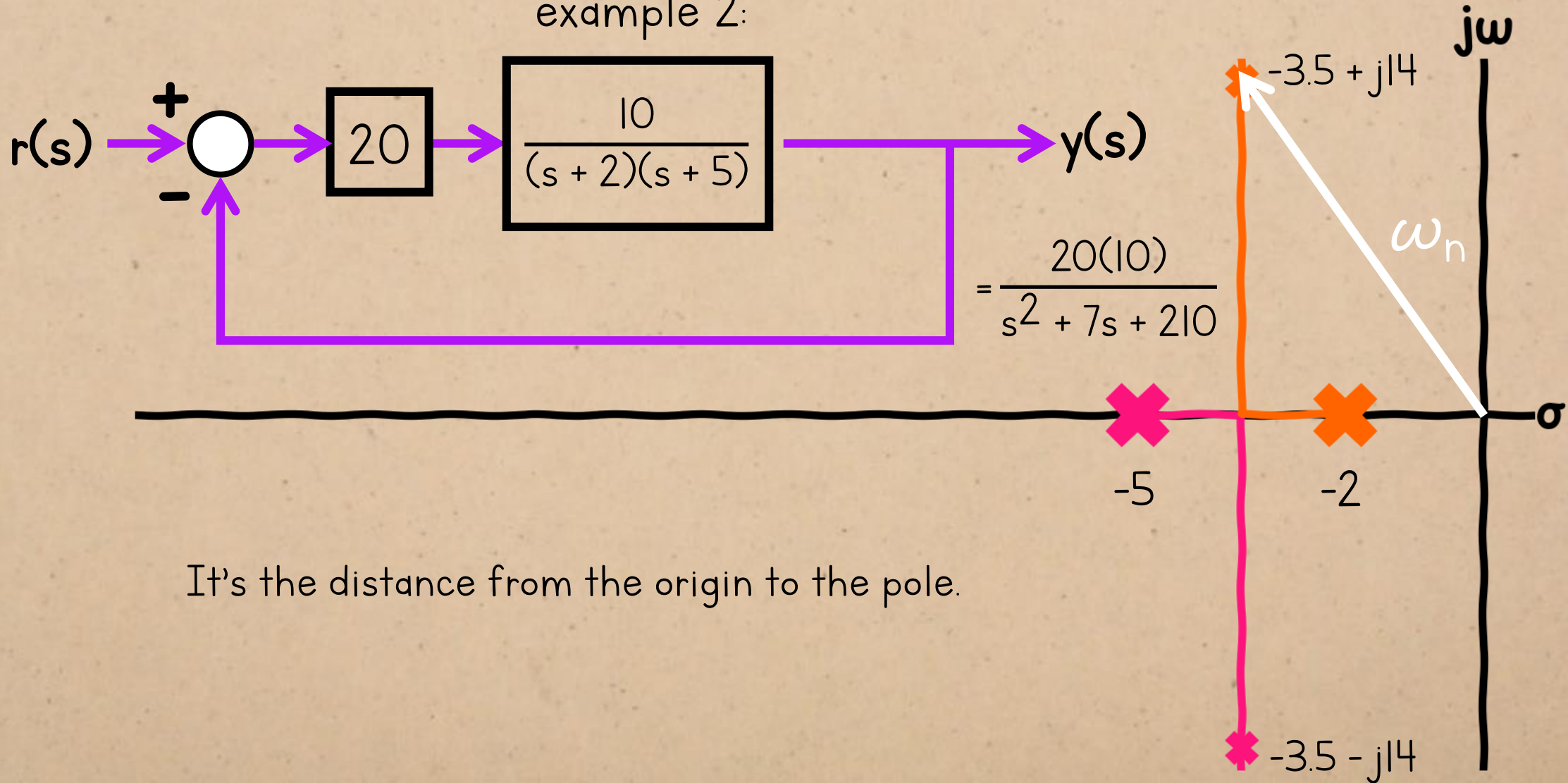
SECOND-ORDER SYSTEM

example 2:



SECOND-ORDER SYSTEM

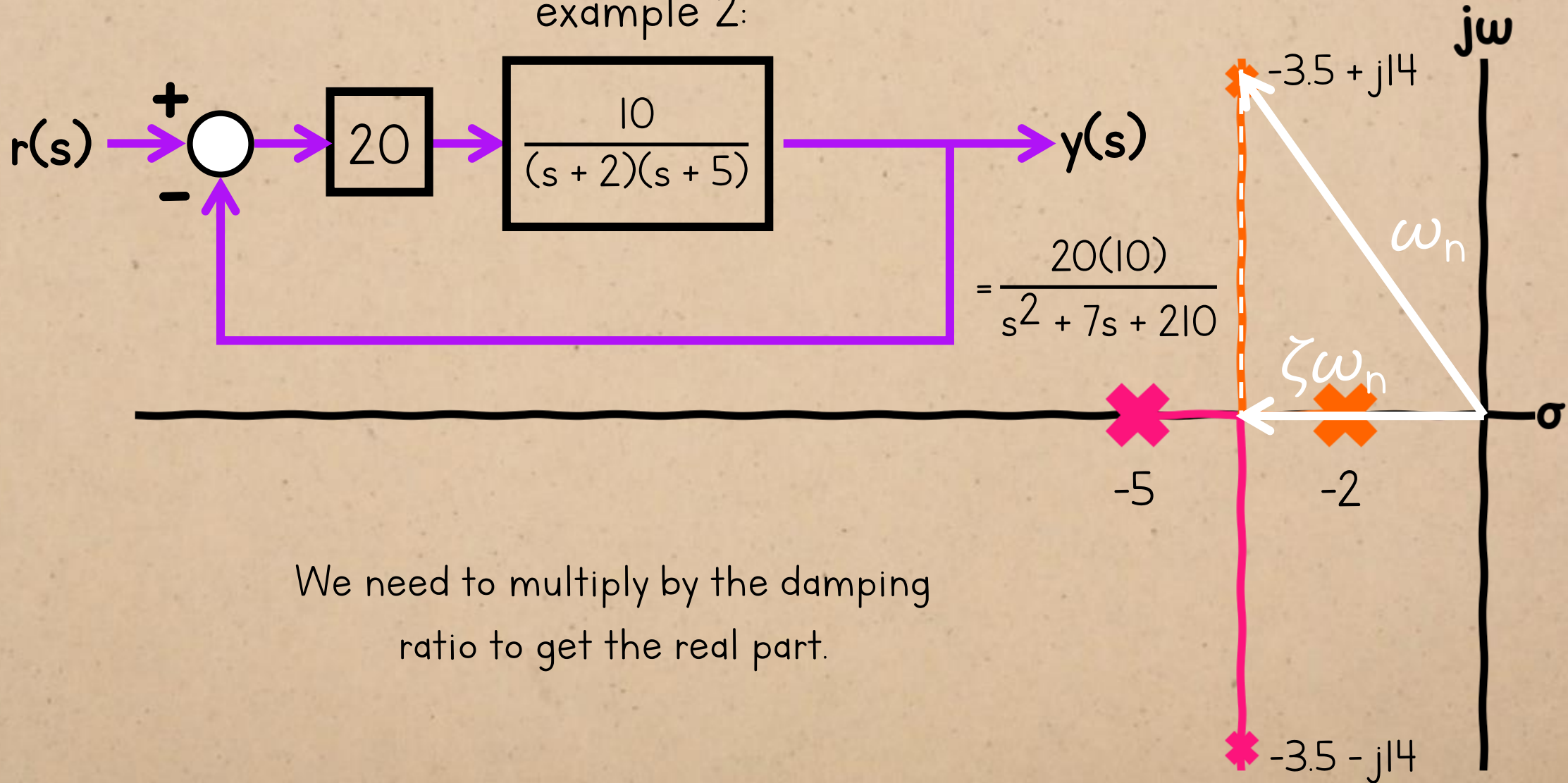
example 2:



It's the distance from the origin to the pole.

SECOND-ORDER SYSTEM

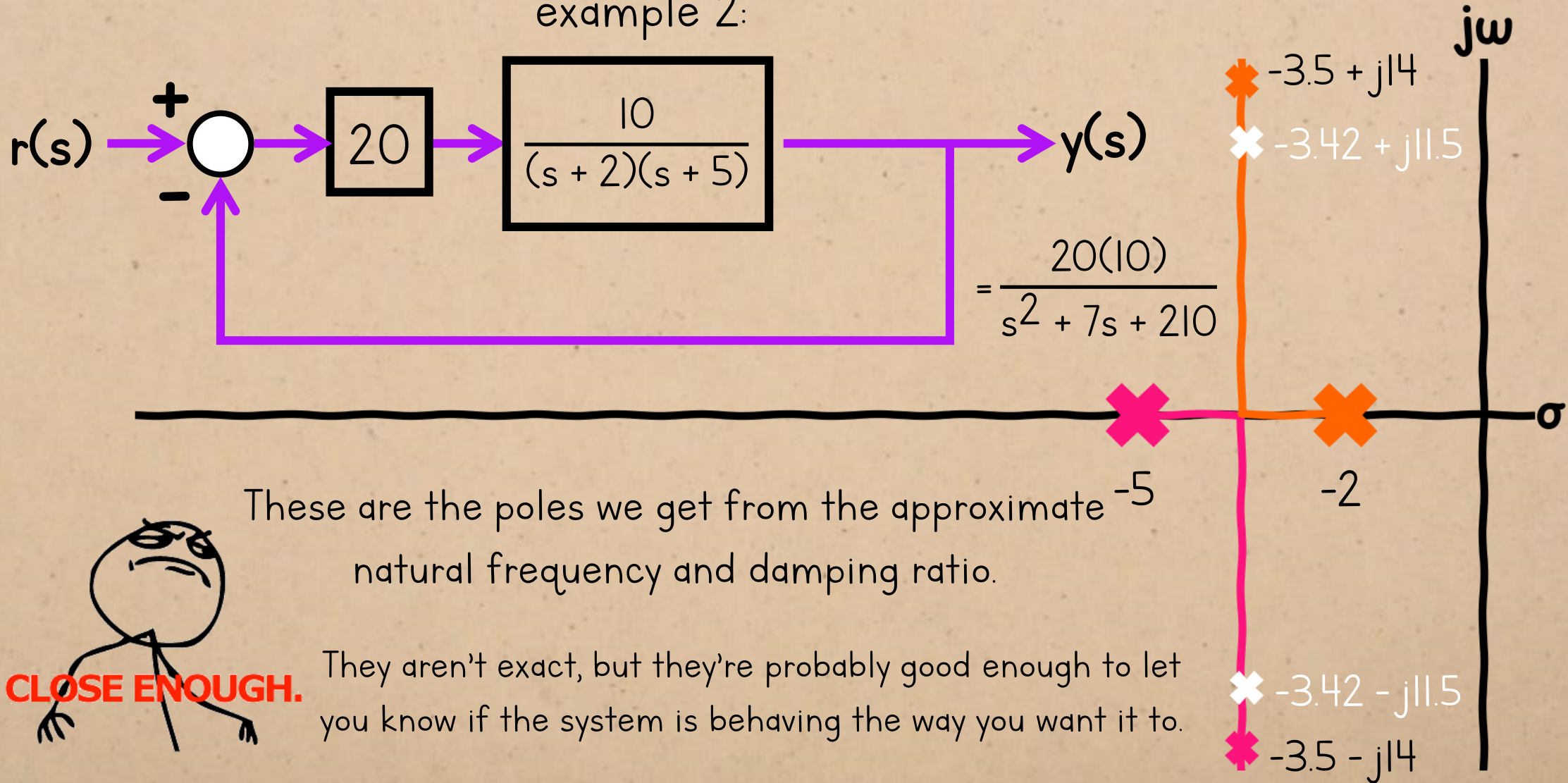
example 2:



We need to multiply by the damping ratio to get the real part.

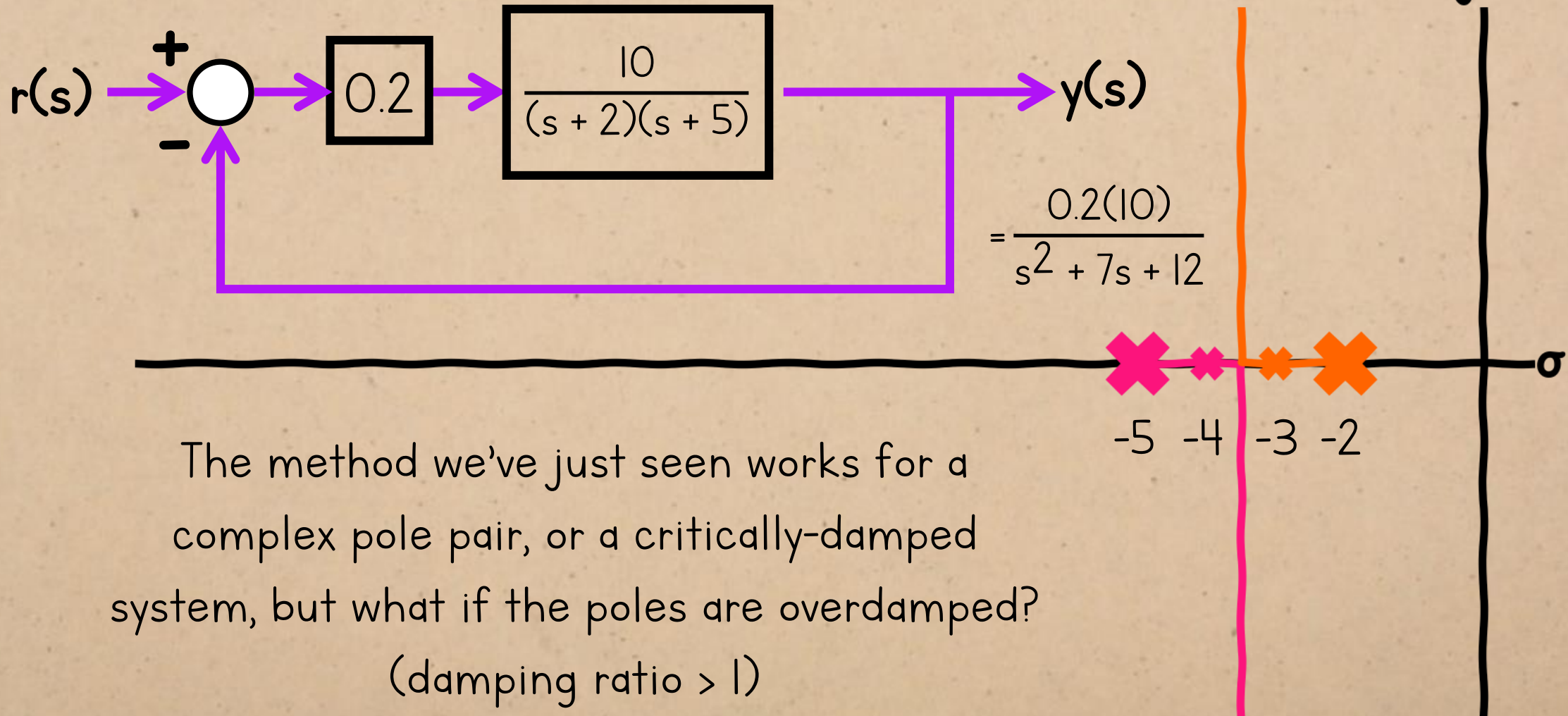
SECOND-ORDER SYSTEM

example 2:



SECOND-ORDER SYSTEM

example 3:



SECOND-ORDER SYSTEM

Don't worry about it.

You're unlikely to come across an overdamped closed-loop system.

THE MATHEMATICAL DETAILS

This is our arbitrary open-loop system:

$$G(s) = \frac{Kp_1p_2}{(s + p_1)(s + p_2)}$$

We need K to be at least 1. Otherwise, the phase margin will be infinite and we won't be able to get the damping ratio from it.

The corresponding closed-loop transfer function is:

$$G_{CL}(s) = \frac{Kp_1p_2}{(s + p_1)(s + p_2) + Kp_1p_2}$$

So the characteristic equation is:

$$s^2 + (p_1 + p_2)s + (K + 1)p_1p_2 = 0$$

From the general second-order transfer function

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_ns + \omega_n^2}$$

this means that $2\zeta\omega_n = p_1 + p_2$ and $\omega_n = \sqrt{(K + 1)p_1p_2}$
(1) (2)

We want to know if it's possible to have $\zeta > 1$ if K is large.

Combining (1) and (2) gives $\zeta = \frac{p_1 + p_2}{2\sqrt{(K + 1)p_1p_2}}$

$$\frac{p_1 + p_2}{2\sqrt{(K + 1)p_1p_2}} > 1$$

$$\therefore p_1 + p_2 > 2\sqrt{(K + 1)p_1p_2}$$

$$\therefore p_1^2 + 2p_1p_2 + p_2^2 > 4(K + 1)p_1p_2$$

$$\therefore K < \frac{1}{4} \frac{p_1}{p_2} + \frac{1}{4} \frac{p_2}{p_1} - \frac{1}{2}$$

So it is possible, but only if the distance between the poles is large.

And in that case, you're probably ok approximating the system as first-order anyway.

TL;DR: don't worry about it.

SUMMARY:

How do you approximate the time constant in the frequency domain?

